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5.1 AUTOMATIC CONTROL SYSTEM AND HISTORICAL DEVELOPMENT

Air Conditioning Automatic Control System

An air conditioning automatic control system or simply a control system, primarily modulates the capacity of the air conditioning equipment to maintain predetermined parameter(s) within an enclosure or for the fluid entering or leaving the equipment to meet the load and climate changes at optimum energy consumption and safe operation. The predetermined parameter or variable to be controlled is called the *controlled variable*. In air conditioning or in HVAC&R, the controlled variable can be temperature, relative humidity, pressure, enthalpy, fluid flow, contaminant concentration, etc.

Because of the variation of the space load and the outdoor climate, a control system is one of the decisive factors for an air conditioning, or an HVAC&R, system to achieve its goal: to effectively control the indoor environmental parameters, provide an adequate amount of outdoor air, be energy-efficient, and provide better security and safety. Today nearly all HVAC&R systems are installed with a control system to provide effective operation and energy conservation.

Historical Development

Early controls for comfort air conditioning systems used in the Capitol building since 1928 were pneumatic controls that included high- and low-limit thermostats in the supply air discharge of the air-handling unit. These controls were used to maintain the desired supply air temperature. Dew-point control of the supply air maintained room humidity within a desirable range. A thermostat in the return-air passage was used to control the air volume flow rate passing through the cooling coil, and a 16-point recorder connected with resistance thermometers was used to record the room temperature at various locations.

Shavit (1995) stated that the use of thermocouples offers the possibility of remote monitoring of the space temperature. In coordination with pneumatic local control systems, the first centralized monitoring system with a central panel and remote set point change was installed in the White House in 1950. The concept of a *building automation system* (BAS) which centralizes the monitoring management of building services was devised and developed in 1950s. The primary improvement of the pneumatic control system occurred in 1963. The vapor and bulb-based sensors were replaced by a remote pneumatic sensor using a bimetal strip and a nozzle flapper to regulate the compressed air pressure between 3 and 15 psi (20.7 and 103.4 kPa).

Shavit also indicated that the first on-line computer was introduced in 1967 in One Main Place in Dallas, Texas. This system had the first set energy conservation software including enthalpy control, demand limit, optimum start/stop, reset according to the zone of highest demand, chiller optimization, and night purge. In 1970, the introduction of all-electronic, solid-state centralized control system was another significant milestone. Solid-state components improved the scanning process and serial transmission as well as reduced many wires in the trunk wiring to a single pair.

The 1973 energy crisis greatly boosted the control industry. Many energy management systems (EMSs) were installed for the purpose of saving energy. The microprocessor-based direct digital control (DDC) was first introduced in 1981 and the DDC unit controller in 1986–1987. By means of the incorporated hardware and software, it tremendously increased the control functions, developed the control logic, and accelerated the data processing and analysis.

Distribution of Energy Management and Control Systems

A building automation system is a centralized monitoring, operation, and management system of the building services in a building including air conditioning (HVAC&R), lighting, fire protection, and security. Only air conditioning control systems and smoke control systems are covered in this handbook.

In the United States, a single-zone, two-position electric control system, as shown in Fig. 5.1, is now widely used in residential and light commercial buildings that adopt DX packaged systems. *Single-zone* means that the load characteristics of the whole conditioned space are similar, it is controlled by one controller, and *two-position* means on-off control. This kind of control system is simple, easy to manage, and low in cost. For a large high-rise building constructed in the 1990s, an energy management and control system (EMCS), as shown in Fig. 5.2, is often the choice. *Variable air volume* indicates that airflow is varied to match the variation of the system load. Today, an EMCS is a microprocessor-based system with direct digital control which optimizes the operation comfortable environment at optimum energy use. Today, an EMCS is an advanced-generation energy management and control system and is also a part of a building automation system.

According to the surveyed results in *Commercial Buildings Characteristics 1992* by EIA, the percentages of the floor area served by the installed EMCS among the total conditioned area in commercial buildings are as follows:

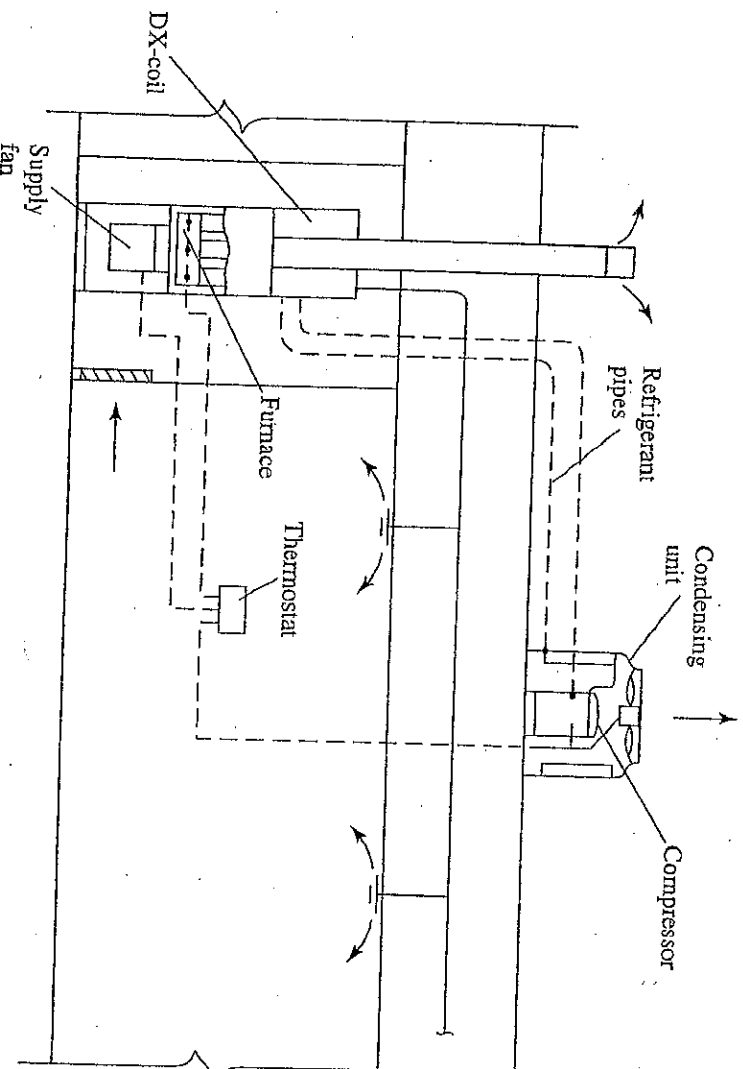


FIGURE 5.1 A single-zone, two-position electric control system.

Year built	Building with EMCS, million ft ²	Percentage of floor space
Before 1970	5431	18
1970-1979	3313	29
1980-1989	4343	36
1990-1992	1236	58

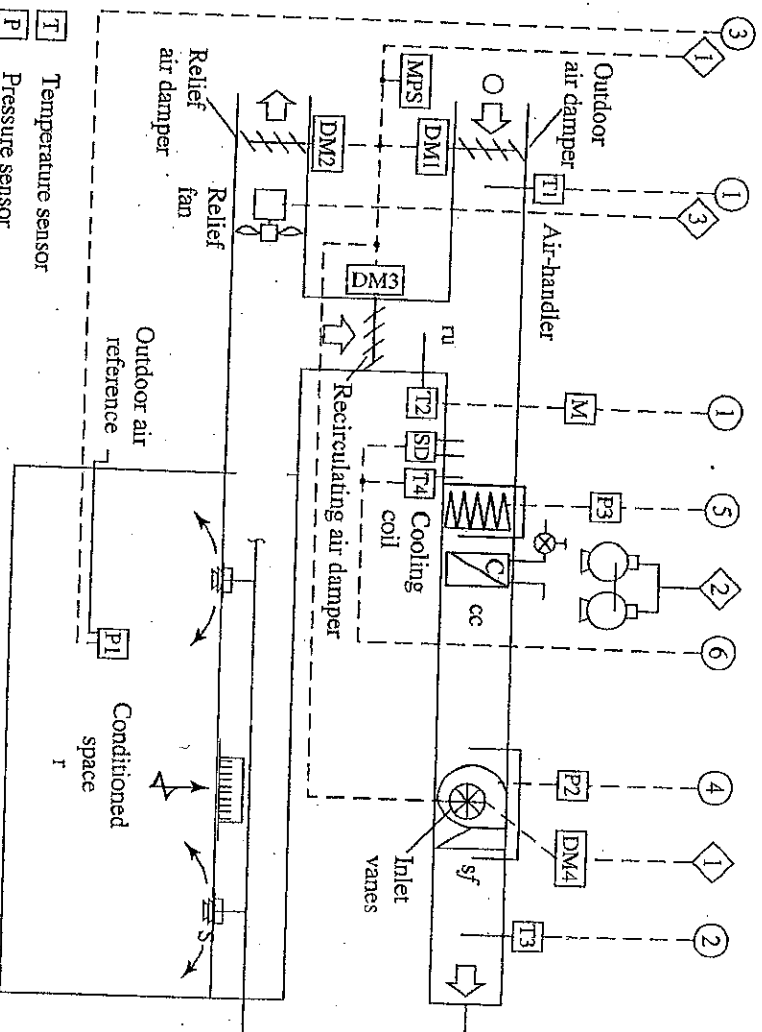


FIGURE 5.2 Control diagram and psychrometric analysis of an energy management and control system for a single-zone, VAV cooling packaged system.

The later the building is constructed and the larger the conditioned area of the commercial building, the greater is the chance to install an energy management and control system (EMS).

5.2 CONTROL LOOP AND CONTROL METHODS

Control Loop

The basic element of a control system is a control loop. A control loop often consists of a sensor (such as T_2 in Fig. 5.2), which senses and measures the controlled variable of recirculating air; a DDC unit controller which compares the sensed input signal with the predetermined condition (the set point) and sends an output signal to actuate the third element, a controlled device or control element (such as a damper or a valve in Fig. 5.2). Modulation, minute adjustments, or on-off control of dampers and valves will change the controlled devices position or operating status, which affects the controlled variable by changing the airflow and water flow, electric power supply, and so on. The controlled variable is thus varied toward the predetermined value, the set point. Air and water equipment that varies the output capacity by changing the opening position of the dampers and valves is called the process plant.

⇒ There are two types of control loops: open and closed. An open-loop system assumes a fixed relationship between the controlled variable and the input signal being received. The sensed variable is not the controlled variable, so there is no feedback. An example of an open-loop system would be a ventilating fan that turns on when the outdoor temperature exceeds a specified set point. The sensed variable is the outdoor air temperature, and the controlled variable is the state of the fan (on or off).

⇒ A closed-loop system depends on sensing the controlled variable to vary the controller output and modulate the controlled device. In Fig. 5.2, temperature sensor T_2 senses the controlled variable, the recirculating air temperature entering the air handler T_m ; the DDC unit controller receives this sensed signal input and produces an output according to the software stored to modulate the fan speed, or the position of the inlet vanes of the supply fan. As the supply fan speed changes, or inlet vanes open and close, the supply air volume flow rate varies and the space temperature T_r and recirculating temperature T_m change accordingly. This change in T_m is sensed again by T_2 and fed back to the controller for further modulation of fan speed or inlet vanes to maintain values of T_m that approach the set points. These components form a closed-loop system.

Figure 5.3 shows a block diagram of this closed-loop system. It shows a secondary input to the controller, such as the outdoor air temperature T_o , which may reset the set point in the controller to provide better and more economical control. The disturbances that affect the controlled variable are load variations and changes in the outdoor weather. After the controller senses the signal feedback, it sends a corrective signal to the controlled device based on the difference between the sensed controlled variable and the set point. Thus, T_m is under continuous comparison and correction.

A control system or its component, control subsystem, used to control the controlled variable(s) in a conditioned space, or within a mechanical device or equipment, may contain only one control loop; or it may contain two or more loops.

Sequence of Operations

The sequence of operations is a description of the sequential order of the functional operations that a control system is supposed to perform, which plans and guides the operation and control of an air conditioning system. For a single-zone variable-air-volume (VAV) cooling packaged system shown in Fig. 5.2, when cold air supply is required during full occupancy, the sequence of operations of cooling mode is as follows:

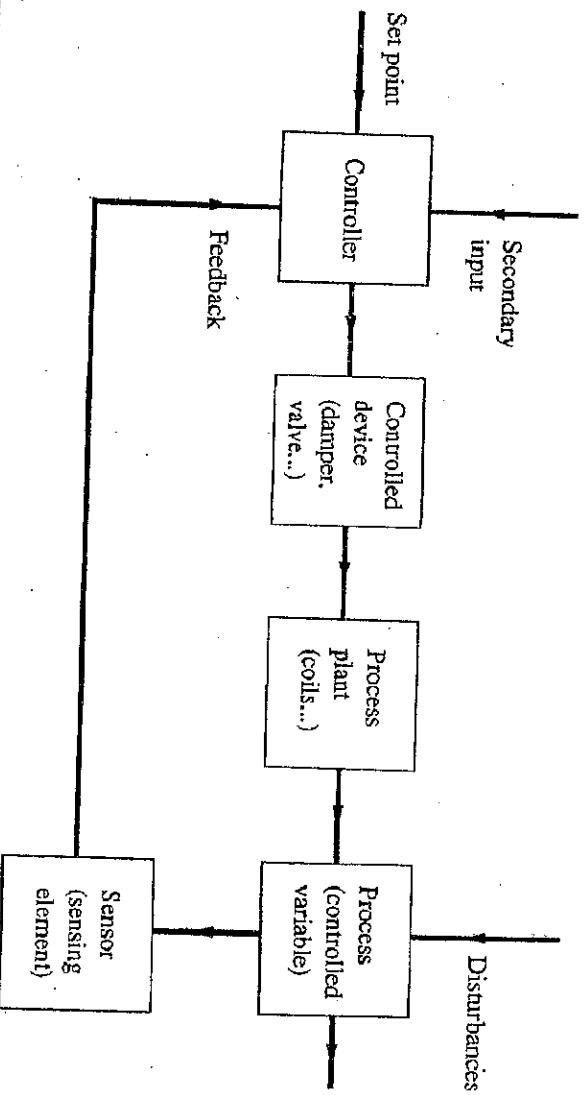


FIGURE 5.3 Block diagram for a closed-loop feedback control system.

- The supply fan is started and stopped by the scheduling software stored in the DDC unit controller. Manual override is possible. When the time schedule puts this system in cooling occupy mode, the microprocessor-based controller goes through a short initiation period, such as a 2-min period. During this period, dampers are driven to fully open, minimum open, and fully closed positions. These determine the effective range of the economizer potentiometer range. The smoke detector in the return air or the low-temperature limit sensor of the mixed air will stop the supply fan if necessary. The supply fan status (on or off) is determined by the pressure differential switch across the fan. When the initiation period is completed, the supply fan is turned on.
- The control system tends to maintain the recirculating temperature T_m at around the cold set point, and it uses the 100 percent all outdoor air free cooling economizer cycle as the first-stage cooling. If the outdoor temperature $T_o \leq T_m$, the outdoor air damper is fully opened and the recirculating air damper closed. If the outdoor air temperature $T_o > T_m$, the outdoor air damper is closed to a minimum position to provide required outdoor ventilation air, and the recirculating damper is fully opened.
- When the initiation period is completed, the supply fan is turned on from its zero speed. If T_m sensed by the temperature sensor T_2 , is at a value above the set point, $T_m > T_{c, set}$, also $T_o \leq T_m$, the speed of the supply fan is gradually increased by the variable-speed drive inverter, resulting in a higher supply volume flow rate. When the fan speed is raised to its upper limit, the supply volume flow rate is then at its maximum value. A still higher space load further raises T_m , and it exceeds the cold set point $T_{c, set}$, and if $T_o \leq T_m$, chilled water starts to flow to the cooling coil to cool the air simultaneously with the outdoor air free cooling, in order to make $T_m \approx T_{c, set}$.
- Only when the outdoor air damper is fully open and the static pressure difference between the space air and outdoor air is greater than a preset value, such as $\Delta p_{s-o} > 0.03$ in. WC (7.5 Pa), will the relief fan be energized. Its speed is modulated to maintain $\Delta p_{s-o} \approx 0.03$ in. WC (7.5 Pa).
- When T_m drops, the chilled water flow to the cooling coil is reduced first, and then the outdoor air free cooling, outdoor air volume flow rate.

Refer to Sec. 21.2 for the details of the sequence of operations of single-zone VAV cooling systems. In the design and operation of an EMCS system, the necessary documentation includes the sequence of operation, control diagrams, specifications, operation, and maintenance manual.

Control Methods

According to the types of control signal and the different kinds of energy used to transmit the signals, as well as whether a software is used during control operation, control methods can be classified as direct digital, pneumatic, electric, and electronic.

⇒ **Analog and Digital.** There are two types of control signals: analog and digital. An analog signal is in the form of a continuous variable. It often uses the magnitude of electric voltage or pneumatic pressure to represent the air temperature. A digital signal is a series of on and off pulses used to transmit information.

A conventional analog controller receives a continuous analog signal, such as a voltage or a pneumatic signal, that is proportional to the magnitude of the sensed variable. The controller compares the signal received from the sensor to the desired value (i.e., the set point) and sends a signal to the actuator in proportion to the difference between the sensed value and the set point.

A digital controller, or microprocessor-based controller, receives an electric signal from sensor(s). It converts the electric signal to digital pulses of different time intervals to represent the signals values. The microprocessor of the digital controller performs the mathematical operations and knowledge processing on these values. The output from the microprocessor can be either in digital form to actuate relays or converted to an analog signal (say, a voltage or a pneumatic pressure) to operate the actuator(s).

⇒ **Direct Digital Control (DDC).** A control system using DDC involves adopting a microprocessor-based digital controller to perform mathematical operations and knowledge processing according to the predetermined control algorithms or computer programs. The key element of DDC compared to analog control is the software and hardware contained in the direct digital controller which expands the control functions tremendously and adopts recently developed control logic. ADDC unit usually has more precise sensors and uses the same type of controlled devices as other control methods.

Figure 1.2 shows an energy management and control system using DDC for an air-handling unit in a typical floor of the NBC Tower, and Fig. 5.2 shows an EMS with DDC for a single-zone VAV system.

⇒ **Pneumatic Control.** In a control system using pneumatic control, compressed air is used to operate the sensors, controllers, and actuators and to transmit the signals. It consists of: a compressed air supply and distribution system, sensors, controllers, and actuators. Figure 5.4 shows a typical pneumatic control system. In Fig. 5.4, a filter is used to remove the dust particles, including submicrometer-size particles, contained in the air. The function of the pressure-reducing valve is to reduce the pressure of compressed air discharged from the air compressor to the required value in the main supply line. The discharged compressed air is usually at a gauge pressure of 18 to 25 psig (124 to 172 kPa·g), and the pressure signal required to actuate the valve or damper actuator is 3 to 13 psig (20.7 to 89.6 kPa·g).

⇒ The advantages of pneumatic control are as follows:

- The compressed air itself is inherently a proportional control signal.
- The cost of modulating actuators is low, especially for large valves and dampers.
- Pneumatic controls require less maintenance and have fewer problems.
- Pneumatic controls are explosionproof.

The disadvantages stem mainly from comparatively fewer control functions, the high cost of sophisticated pneumatic controllers, and the comparatively higher first cost of a clean and dry, compressed air supply for small projects.

⇒ **Electric or Electronic Control.** Both types of control use electric energy as the energy source for the sensors and controllers. A control system using electric control often offers two-position on-off control, as shown in Fig. 5.1. Switches, relays, contactors, and electromechanical devices are

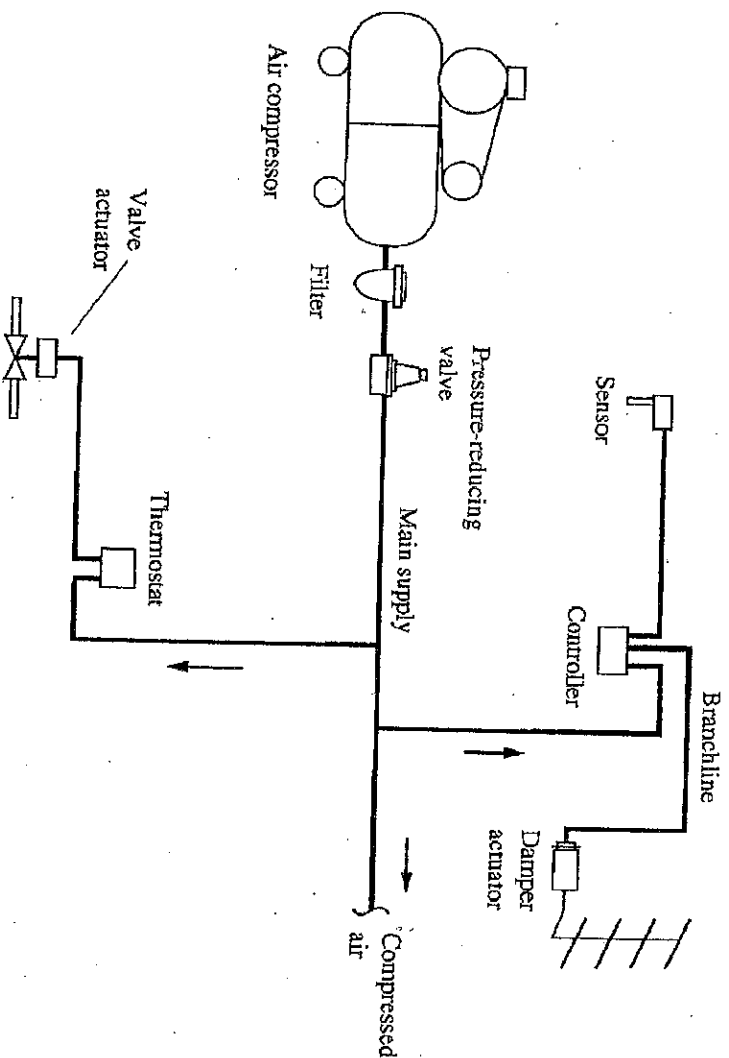


FIGURE 5.4 A typical pneumatic control system.

system components for electric control systems. They are generally used for low-cost, small, and simpler control systems. In addition to the switches and relays, a control system using electronic control has transistors, diodes, capacitors, and printed-circuit boards as system components. Electronic control systems always have more accurate sensors and solid-state controllers with sophisticated functions and can be easily interfaced with the building automation system. Electronic control has a faster response and more accurate processing of data than electric control systems. Electronic control systems cost more and need skilled personnel for maintenance and troubleshooting.

Comparison of Control Methods

Because of the increasing demand for more complicated controllers to satisfy the needs of better indoor environmental control, satisfactory indoor air quality, improved energy saving, lower cost, and greater reliability, the recent trend is to use EMCS with direct digital control for more demanding and large projects. A modern DDC system consists of electronic sensors, microprocessor-based controllers incorporated with electronic components, and electronic or pneumatic actuators. It is often more effective and cheaper to operate a large pneumatic actuator than to use a large electronic actuator in a DDC system.

An EMCS using DDC offers the following advantages:

1. Flexibility of providing required control functions and the ability to coordinate multiple functions directly from complicated software programs (it can even mimic a human expert within a certain knowledge domain and offer artificial intelligence)
2. More precise and faster-response control actions provided by the microprocessor-based controller
3. The possibility of using high-level self-checking and self-tuning system components, which increases the system reliability

One of the main disadvantages of DDC is a still-higher first cost. However, the cost per control action is comparatively lower. As more and more DDC systems are installed, DDC will become familiar to us. The cost of the microprocessor-based controllers and other DDC system components will drop further in the future. In the 1990s, the trend in HVAC&R control for new and retrofit projects is to use EMCS with DDC systems except for small projects.

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5.3 CONTROL MODES

Control modes describe how the corrective action of the controller takes place as well as its effect on the controlled variable. For applications in HVAC&R, control modes can be classified as two-position, step, floating, proportional, proportional-integral, and proportional-integral-derivative.

Before we discuss the control modes in detail, the term *lag*, or *time lag*, should be introduced. According to ASHRAE terminology, lag is (1) the time delay required for the sensing element to reach equilibrium with the controlled variable; or (2) any retardation of an output with respect to the causal input, including the delay, because of the transport of material or the propagation of a signal.

Two-Position Control

In two-position control, the controller controls the final control element at one of two positions: maximum or minimum (except during the short period when it changes position). Examples of two-position control include starting and stopping the motor of a fan, pump, or compressor and turning on or off an electric heater. Sometimes it is called on-off control.

Figure 5.5 shows a two-position control mode for an electric heater installed in a branch duct. In the middle diagram, the ordinate indicates the controlled variable, the space temperature T_s ; in the lower diagram, the ordinate denotes the output capacity of the final control element, the electric heater. If the controller turns on the electric heater when the sensor senses a space temperature 69.5°F (20.8°C), and turns off at 70.5°F (21.4°C), the result is a cyclic operation of the electric heater and the rise and fall of T_s toward the two positions of 69.5°F (20.8°C) and 70.5°F (21.4°C). The thermal storage effect of the electric heater, the branch duct, the building envelope that surrounds the space, and the sensor itself will have a time lag effect on T_s . The rise of T_s is a convex curve with an overshoot higher than 70.5°F (21.4°C), and the drop of T_s is a concave curve with an undershoot lower than 69.5°F (20.8°C). The difference between the two points, on and off, is called the *differential*.

If the heating capacity of the electric heater that results from an on-and-off cyclic operation is represented by $Q_{e,n}$, and if the actual space heating load $Q_h < \frac{1}{2}Q_{e,n}$, the slope of the rising T_s curve will then be greater than the slope of the falling T_s curve. As a consequence, the "on" period will be shorter than the "off" period. Such a cyclic T_s curve is shown in the upper left corner of Fig. 5.5. If $Q_h > \frac{1}{2}Q_{e,n}$, the condition will be reversed (see the upper right corner of Fig. 5.5). The two-position control varies the ratio of on and off periods to meet any variation in the space heating load.

To reduce the overshoot and undershoot of the controlled variable in a two-position control mode, a modification called *timed two-position control* has been developed. A small heating element attached to the temperature sensor is energized during on periods. This additional heating effect on the sensor shortens the on timing. During off periods, the heating element is deenergized.

The differential of two-position control, the overshoot, and the undershoot all result in a greater fluctuation of the controlled variable. A suitable differential is always desirable in two-position control in order to prevent very short cycling, which causes hunting, a phenomenon of short cycling of the controlled variable. Two-position control is not suitable for precise control of the controlled variable, but it is often used for status control, such as opening or closing a damper, turning a small single piece of equipment on or off for capacity control, etc., and for lower-cost control systems.

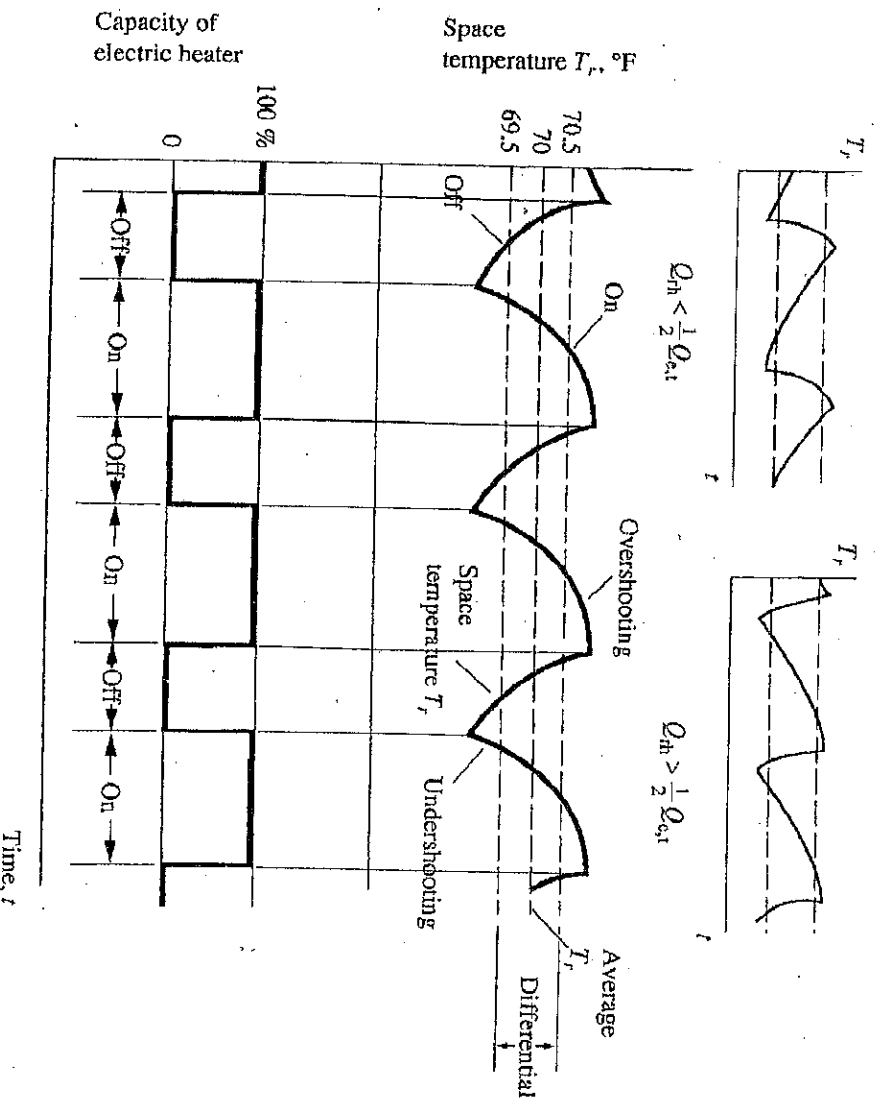


FIGURE 5.5 A typical two-position control mode.

Step Control and Modulating Control

In step control, the controller operates the relays or switches in sequence to vary the output capacity of the process plant in steps or stages. The greater the deviation of the controlled variable from the set point and the faster the rate of change of the controlled variable, the higher will be the output capacity of the process plant. In modulating control, the controller activates the control device continuously and thus the change of the output capacity gradually.

During the earlier stage of step control, there must be a differential of controlled variable between two on or two off points of a particular piece of equipment of the process plant. The result is a greater fluctuation of the controlled variable. In the DDC controller-activated new generation of step control, the software of the DDC unit determines how fast each capacity step is added or subtracted according to the deviation from the set point and the rate of change of the controlled variable.

Figure 5.6 shows step control of the cooling capacity of refrigeration compressors using DDC to maintain the discharge air temperature T_{dis} within predetermined limits in a packaged unit. When the air economizer alone can no longer balance the DX coils load, and T_{dis} floats to the upper limit of the control band, point 1, the first stage of cooling is energized. If the cycling and the continuously energized first-stage cooling capacity still cannot balance the coils load, then T_{dis} continues to rise until it reaches point 2, which is 1°F (0.56°C) higher than the upper limit of the control band. The first-stage cooling is then locked on, and the second stage is energized to cycling. When T_{dis} rises to point 2, all the present cooling stages will be locked on, and an additional cooling stage is added for cycling. When T_{dis} drops to point 7, the current cycling cooling stage will be deenergized, and the one cooling stage next to the deenergized stage will be cycling. It always needs a time delay, say at least 4 min, to turn on or off again in order to prevent hunting and other damage.

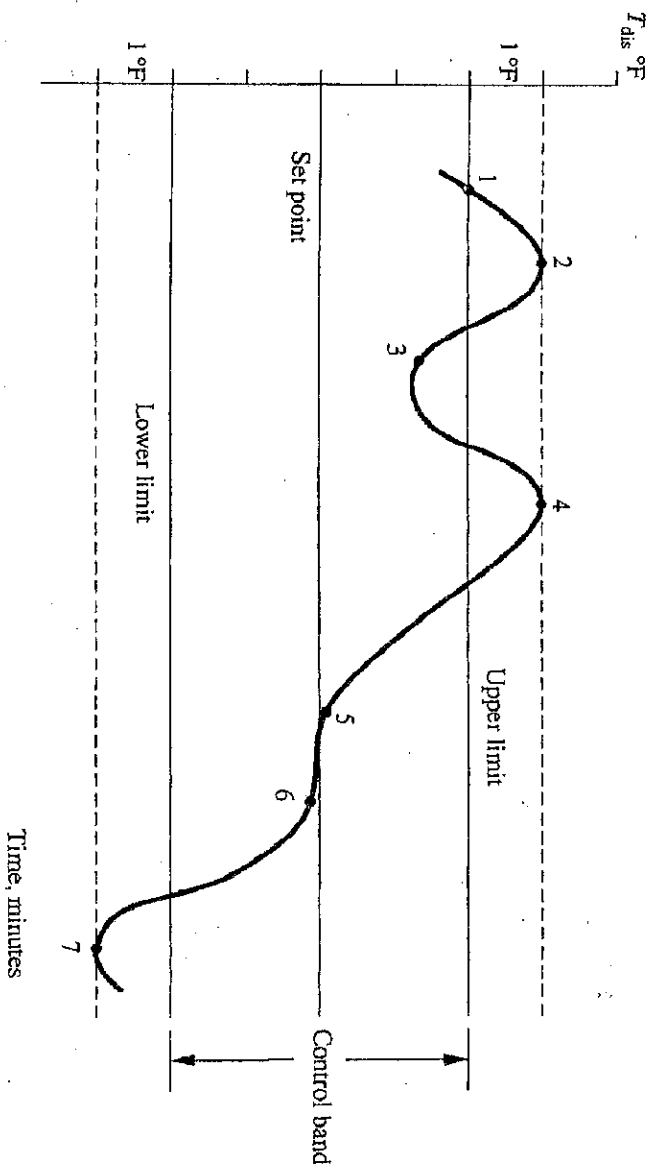


FIGURE 5.6 DDC-activated discharge air temperature control for a VAV rooftop packaged system.

Step control is another kind of on-and-off control having smaller varying capacity and thus a fluctuating controlled variable. It has been widely used for the capacity control of refrigeration compressors and the electric heaters.

Floating Control

In a floating control mode, the controller moves the control device by means of the actuator toward the set point only when the control point is out of the differential, or dead band, as shown in Fig. 5.7. A control point is the actual value of the controlled variable at a certain time instant. Figure 5.7 represents a duct static pressure control system using a floating control mode by opening and closing the inlet vanes of a supply fan. The control device—inlet vanes—can be moved in either an opening or a closing direction depending on whether the control point is over the upper limit of the differential or under the lower limit. In Fig. 5.7, when the controlled variable, the duct static pressure Δp_s , is above the upper limit of the differential, the controller then closes the inlet vanes. If Δp_s is below the lower limit of the differential, the controller opens the inlet vanes.

A floating control mode is more suitable for control systems with a minimal lag between the sensor and the control medium. A control medium is the medium in which the controlled variable exists.

Proportional Control

In a proportional control mode, the controller moves the controlled device to a position such that the change in its output capacity is proportional to the deviation of the controlled variable from the set point. The position of the controlled device is linearly proportional to the magnitude of the controlled variable. Figure 5.8 represents the control of the space temperature through the modulation of a two-way valve of a cooling coil using proportional control. In Fig. 5.8, the controlled variable is the space temperature T_s , and the controlled device is the valve.

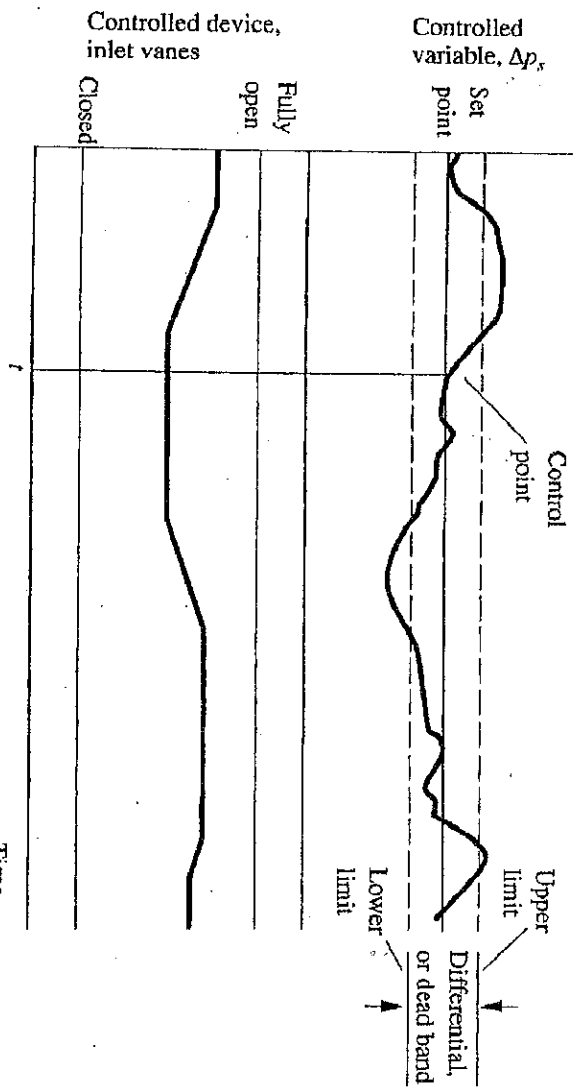


FIGURE 5.7 Floating control mode.

In a proportional control mode, the throttling range is the change in the controlled variable when the controller moves the controlled device from the position of maximum output to the position of minimum output. The controlled variable's range of values that will move the proportional controller through its operating range is called the *proportional band*. In a proportional control system, the throttling range is equal to the proportional band.

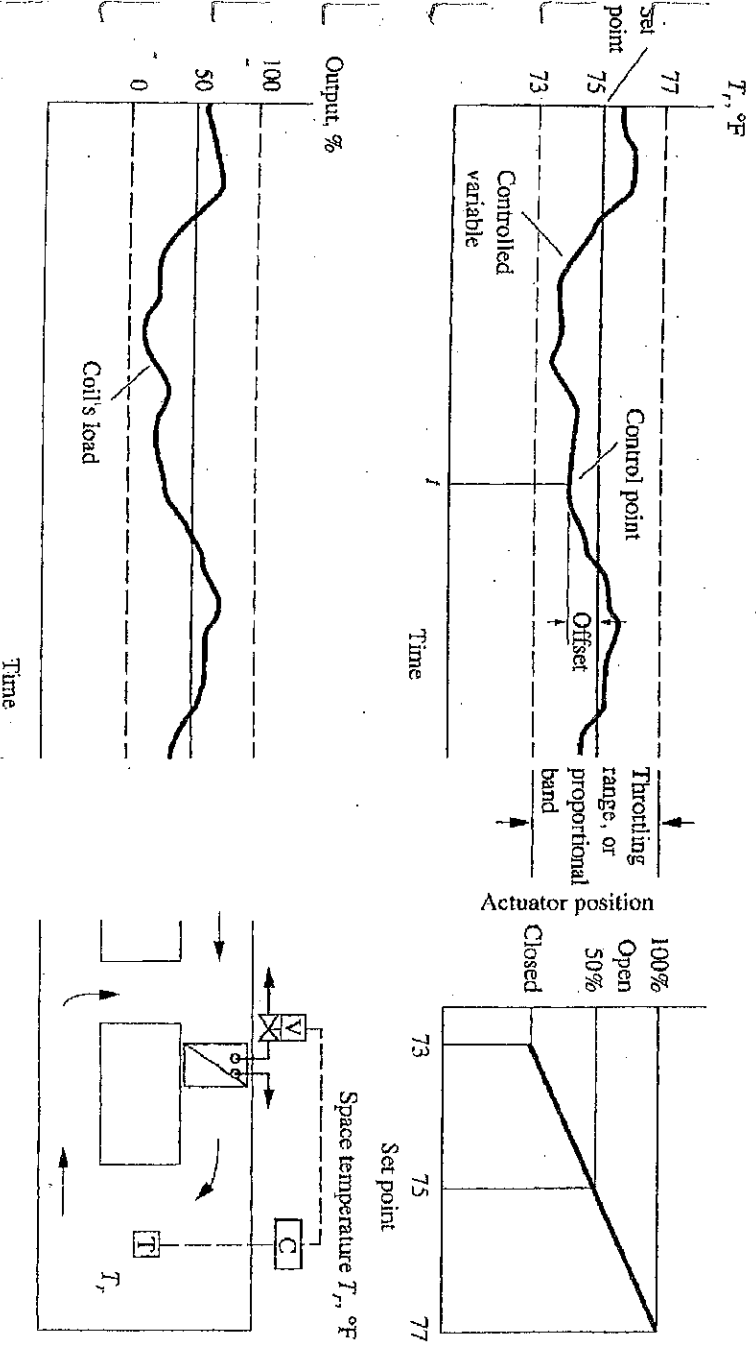


FIGURE 5.8 Proportional control mode.

The set point is the desired value of the controlled variable, or the desired control point that the controller seeks to achieve. The difference between the control point and the set point is called the *offset*, or *deviation*. In proportional control, when the controlled variable is at the bottom line of the throttling range, the controller will position the actuator at the closed position. At the set point, the actuator will be at 50 percent of the open position. When the controlled variable is at the top of the throttling range, the actuator will be at 100 percent of the open position.

In a proportional control mode, since the output signal V of the controller is proportional to the deviation of the control point from the set point, their relationship can be expressed as:

$$V = K_p e + M \quad (5.1)$$

where K_p = proportional gain, proportional to $1/(\text{throttling range})$

e = error signal, i.e., deviation or offset

M = output value when deviation is zero (usually, output value at middle of output range of controller)

For space and discharge air temperature control using proportional control, the offset is directly proportional to the space load and the coil load. The space or discharge temperature T , $^{\circ}\text{F}$ ($^{\circ}\text{C}$), can be calculated as:

$$T = R_{\text{load}} T_{lr} + T_{\text{min}} \quad (5.2)$$

where T_{lr} = throttling range, expressed in terms of space or discharge temperature, $^{\circ}\text{F}$ ($^{\circ}\text{C}$)

T_{min} = space and discharge temperature when space load or coil load is zero, $^{\circ}\text{F}$ ($^{\circ}\text{C}$)

In Eq. (5.2), R_{load} represents the load ratio of the space load or coil load, which is dimensionless and can be evaluated as

$$R_{\text{load}} = \frac{\text{actual load}}{\text{design load}} \quad (5.3)$$

The actual load and design load must be in the same units, such as Btu/h (W).

For a proportional control mode, a certain degree of offset, or deviation, is inherent. The position of the controlled device is a function of the offset. Only when offset exists will the controller position the actuator and the valve or damper at a position greater or smaller than 50 percent open. The throttling range is primarily determined by the HVAC&R system characteristics and cannot be changed after the system is designed. A proportional control mode is suitable for an HVAC&R system that has a large thermal capacitance, resulting in a slow response and stable system and allowing comparatively narrower throttling range and thus a smaller offset. Fast-reacting systems need a large throttling range to avoid instability and short cycling, or hunting.

Proportional plus Integral (PI) Control

In a proportional plus integral control mode, a second component, the integral term, is added to the proportional action to eliminate the offset. The output of the controller can thus be expressed as

$$V = K_p e + K_i \int e \, dt + M \quad (5.4)$$

where K_i = integral gain and t = time.

In Eq. (5.4), the second term on the right-hand side (the integral term) indicates that (1) the error, or offset, is measured at regular time intervals and (2) the product of the sum of these measurements and K_i is added to the output of the controller to eliminate the offset. The longer the offset exists, the greater the response of the controller. Such a control action is equivalent to resetting the set point in order to increase the controller output to eliminate the offset. As such, PI control is sometimes called proportional plus reset control. Figure 5.9a shows the variations of a controlled

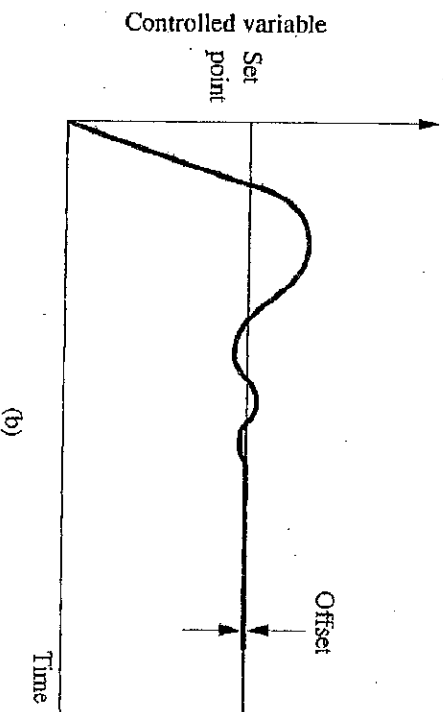
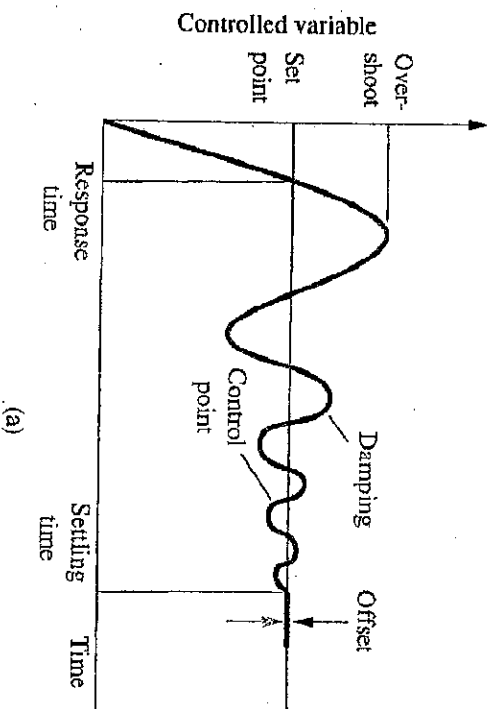


FIGURE 5.9 Proportional-integral and proportional-integral-derivative control modes. (a) Proportional-integral; (b) proportional-integral-derivative.

variable for a proportional-integral control mode. In PI control, proportional-integral control signals are additive.

For PI control, the controlled variable does not have any offset once it has achieved a stable condition, except due to any inaccuracy of the instrumentation and measurement. Proper selection of the proportional gain K_p and integral gain K_i as well as proper tuning is important for system stability and control accuracy. PI control may be applied to fast-acting control systems with a greater throttling range setting at the controller for better system stability, e.g., discharge air temperature, discharge chilled water temperature, and duct static pressure control systems.

Proportional-Integral-Derivative (PID) Control

A PID control mode has additional control action added to the PI controller: a derivative function that opposes to any change and is proportional to the rate of change. The output of such a controller can be described by the following equation:

$$V = K_p e + K_i \int e \, dt + K \frac{de}{dt} + M \quad (5.5)$$

where K is the derivative gain. The effect of adding the derivative function $K \, d\epsilon/dt$ is that the quicker the control point changes, the greater the corrective action provided by the derivative function. Figure 5.9b shows the variation of the controlled variable for a PID control mode. As with a PI control mode, PID control mode also has no offset once the controlled variable has reached a stable condition, except due to instrument inaccuracy.

Compared to the PI control mode, the PID control mode, which is a combination of proportional, integral, and derivative actions, exhibits faster corrective action and a smaller overshoot and undershoot following an offset and a change of the controlled variable. The controlled variable is brought to the required set point in a shorter time. However, it is more difficult to determine properly three constants, or gains (K_p , K_i , and K_d).

Compensation Control, or Reset

Compensation control, or reset, is a type of control mode in which a compensation sensor is generally used to reset a main sensor to compensate for a variable change sensed by the compensation sensor. The purpose is to achieve operation that is more effective, energy-efficient, or both.

In the design of a reset mode, the first things to decide are the control point at which the main sensor will be reset and the variable to be sensed by the compensation sensor. The main sensor, which senses the mixed air temperature in the air-handling unit, is usually reset by a compensation sensor that senses the outdoor temperature, as shown in Fig. 5.10.

Another task to decide on is the relationship between the variables sensed by the main and compensation sensors—the reset schedule. Many reset schedules have a different relationship between these two sensors at various stages. For example, in Fig. 5.10, when the outdoor temperature T_o is less than 30°F (-1.1°C), it is within the range of stage I. No matter what the magnitude of T_o is, the set point of the mixed temperature T_m is 65°F (18.3°C). When $30^\circ\text{F} \leq T_o \leq 95^\circ\text{F}$ ($-1.1^\circ\text{C} \leq T_o \leq 35^\circ\text{C}$), it is in stage II. The linear relationship between the temperatures sensed by the main and compensation sensors in Fig. 5.10 can be expressed by the following equation:

$$T_m = 65 - \frac{65 - 55}{95 - 30} (T_o - 30) \quad (5.6)$$

When $T_o > 95^\circ\text{F}$ (35°C), the set point of the mixed temperature is always 55°F (12.8°C).

Compensation control modes have been widely adopted to reset space temperature, discharged air temperatures from air-handling units or packaged units, or water discharge temperatures from central plants.

Applications of Various Control Modes

Selection of a suitable control mode depends on

- Operating characteristics
- Process or system characteristics, such as whether the thermal capacitance should be taken into consideration
- Characteristics of load changes

If a simpler control mode can meet the requirements (say, two-position control versus PID control), the simpler control mode is always the first choice. Except for two-position and step control, all the other control modes are modulation controls. Modulation control is a control mode that is capable of increasing or decreasing a variable according to the deviation from the required value in small increments continuously. Also note that there is a significant difference between the two-position control, step control, floating control, and proportional control modes with an offset and the offset free PI and PID controls.

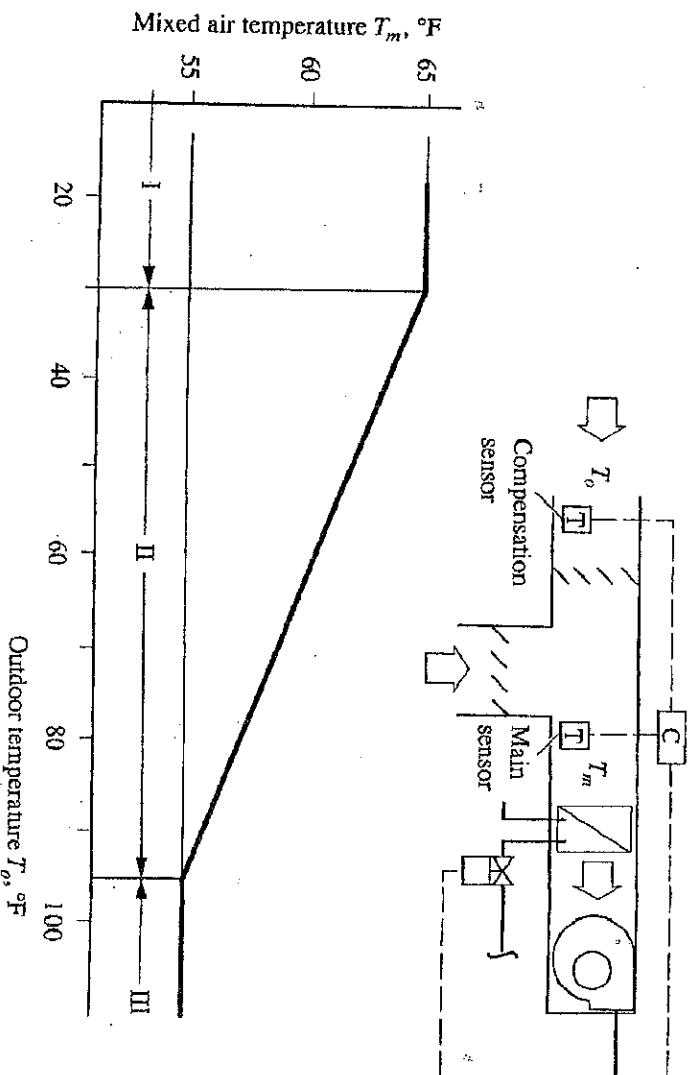


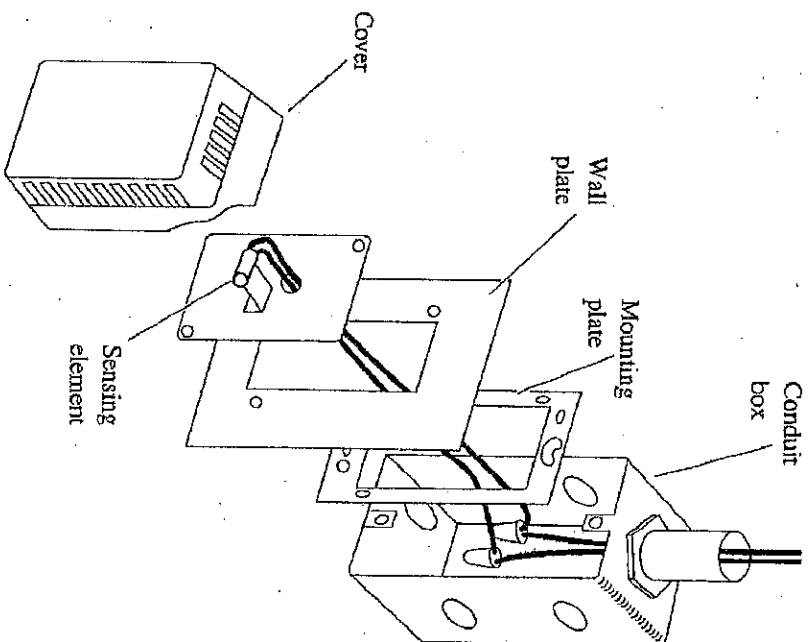
FIGURE 5.10 Reset control and schedule.

For HVAC&R processes, PI control can satisfy most of the requirements, and it is most widely used. Microprocessor-based PID control is a very powerful tool. PID control mode is more appropriate for fast-acting duct static pressure control and airflow control, and it is recommended to set the controller with a large proportional band for control system stability, a slow reset to eliminate deviation, and a derivative action to provide a quick response.

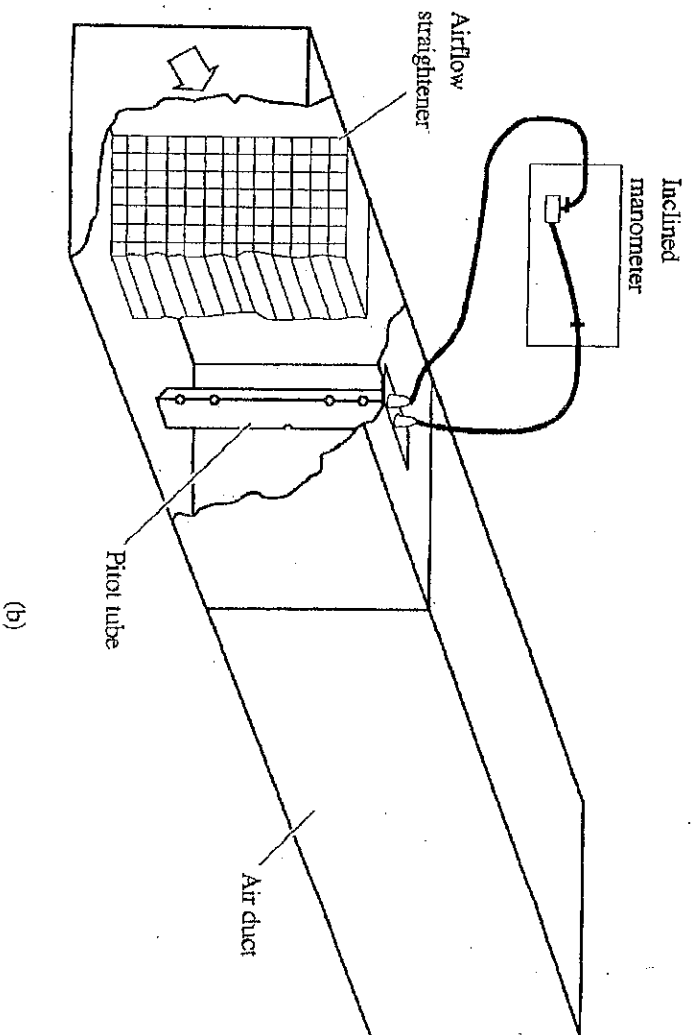
5.4 SENSORS AND TRANSDUCERS

A sensor is a device that acts as a component in a control system to detect and measure the controlled variable and to send a signal to the controller. A sensor consists of a sensing element and accessories, as shown in Fig. 5.11. The term *sensing element* often refers to that part of the sensor that actually senses the controlled variable. In HVAC&R systems, the most widely used sensors are temperature sensors, humidity sensors, pressure sensors, and flow sensors. Recently, CO₂ sensors, air quality sensors, and occupancy sensors are being used in many new and retrofit projects. Electronic sensors that send electric signals to electronic controllers, can also be used for the DDC units, and the current trend is to use solid-state miniaturizing sensing elements. Electric output from the sensors is usually expressed in 0 to 10 V dc or 4 to 20 mA. In the selection of sensors, accuracy, sensitivity of response, reliability, long-term stability or drift, required calibration intervals, maintainability, and especially the possibility of contamination by dust particles due to contact with the control medium should be considered. Drift is a kind of offset. ASHRAE's *Terminology* defines *drift* as “change in an output-input relationship over time with the change unrelated to input, environment, or load.” A calibration period of 1 year and longer is often considered acceptable.

Location of the sensor affects the sensor output directly. A sensor should be located in the critical area where the controlled variable needs to be maintained within required limits. The sensing element of an air sensor should be well exposed to the air, so that air can flow through the sensor without obstruction. Air sensors should not be affected directly by the supply airstream nor should space air sensors be affected by the outdoor airstream. Sensors should be shielded from radiation



(a)



(b)

FIGURE 5.11 (a) A temperature sensor and (b) a typical pitot tube flow-measuring station.

and mounted firmly on a structural member or duct wall, free from vibration. If a space air sensor must be located on a concrete column, thermal insulation should be provided between the sensor and the column.

Temperature Sensors

Temperature sensors fall into two categories: those that produce mechanical signals and those that emit electric signals. Bimetal and rod-and-tube sensors that use sensing elements to produce a mechanical displacement during a sensed temperature change either open or close an electric circuit in an electric control system, as shown in Fig. 5.1, or adjust the throttling pressure by means of a bleeding nozzle in a pneumatic control system. For a sealed bellows sensor, a change in temperature causes a change in the pressure of a liquid in a remote bulb. The expansion and contraction of vapor then move the mechanism of the controller.

Temperature sensors that produce electric signals, as shown in Fig. 5.11a, are the same as the sensors for temperature measurement and indication mentioned in Sec. 2.3.

In addition to resistance temperature detectors (RTDs) and thermistors, sensors sometimes use thermocouples. A thermocouple uses wires of two dissimilar metals, such as copper and constantan, or iron and nickel, connected at two junctions, to generate an electromotive force between the junctions that is directly proportional to the temperature difference between them. One of the junctions is kept at constant temperature and is called the cold junction. Various systems have been developed to maintain the cold junction at a constant temperature and to provide compensation if the cold junction is not at 32°F (0°C). This task makes the use of thermocouples more complicated and expensive. The electromotive force produced between the two junctions can be used as the signal input to a controller.

Bimetal and rod-and-tube temperature sensors are simple and low in cost. However, they cannot provide temperature indication and electric signals for DDC and are often used in electric control systems. Platinum and nickel RTDs are stable, reliable, and accurate. They are very expensive compared to thermistors and need calibration to compensate for the effects of having external leading wires. RTDs are widely adopted in DDC for commercial applications. For a project that needs precise temperature control, RTD is often the choice. High-quality thermistors can provide stable, reliable, and interchangeable temperature sensors, and are also widely used in many commercial applications. Petze (1986) reported that some high-quality thermistors exhibited better than $\pm 0.002^\circ\text{F}$ ($\pm 0.001^\circ\text{C}$) stability for a 2-year period.

A typical space air temperature sensor has the following characteristics:

Sensing element	Platinum film element, 3000 Ω
	Positive temperature coefficient, 4.8 $\Omega/^\circ\text{F}$ (8.64 $\Omega/^\circ\text{C}$)
Operating range	60 to 90°F (15.6 to 32.2°C)

Humidity Sensors

As mentioned in Sec. 2.8, humidity sensors fall into two categories: mechanical and electronic. When the same humidity sensor is used for both monitoring and DDC, the ion-exchange resistance-type, also called the *bulk polymer resistance-type*, and capacitance-type humidity sensors are often adopted. Humidity sensors usually have different accuracy at very low, 20 to 80 percent mid-range, and high relative humidities. The bulk polymer resistance humidity sensor is not accurate at low relative humidities and generally provides stable and accurate readings within a range of 30 to 90 percent relative humidity. Its performance is affected by the air contamination. Capacitance humidity sensors are accurate at 10 to 80 percent relative humidity. However, they become unstable at high relative humidities.

In an EMCS, an electronic humidity sensing element measures the relative humidity and sends an electric signal to the DDC unit. A typical space air humidity sensor has the following specifications:

Power source	24 V ac
Humidity range (active)	10 to 80 percent
Nominal output range	0 to 5 V dc
Accuracy at 70°F (21.1°C)	± 3 percent at 10 to 60 percent range ± 4 percent at 60 to 80 percent range
Speed of response	8 min (90 percent of response time)
Mounting	Wall mount

Pressure Sensors

A pressure sensor usually senses the difference in pressure between the controlled medium (air and water) and a reference pressure; or the pressure differential across two points, such as the pressure differential across a filter. The reference pressure may be an absolute vacuum, atmospheric pressure, or the pressure at any adjacent point. Output signals from the pressure sensors may be electric or pneumatic, and analog or binary.

Pressure sensors used for HVAC&R systems can be divided into high-pressure and low-pressure sensors. High-pressure sensors measure in pound per square inch or feet WC (kilo pascals), and low-pressure sensors measure in inches WC (pascals). The sensing elements for high-pressure sensors are usually Bourdon tube, bellows, and sometimes diaphragms. For low-pressure air sensors, large diaphragms or flexible metal bellows are usually used.

To measure the duct static pressure, if a long section of straight duct of a length greater than 10 ft (3 m) is available, a single-point, pitot-tube type of duct static pressure sensor can be used. Otherwise, a multipoint pitot-tube array or flow-measuring station with airflow straighteners should be used. The small hole used to measure static pressure should never be directly opposite to an airstream with a velocity pressure that can affect the reading. A reference pressure should be picked up at a point with low air velocity outside the duct, at a point served by the same air system, or in the ceiling plenum.

In a DDC system, low air pressure is often sensed by measuring the capacitance of two diaphragms; one is allowed to move toward or away from the fixed one, depending on the pressure on two sides of the diaphragm. An electric circuit is used to convert the capacitance to a voltage or milliampere signal. The other method used to sense the low air pressure in DDC systems works because the air pressure differential compresses or stretches a diaphragm as well as a strain gauge. The change of the resistance of the gauge is detected and amplified, and this electric signal in the form of voltage or milliamperes is sent to the controller.

A space air pressure sensor having an operating range of + 0.1 to - 0.1 in WC (+ 25 Pa to - 25 Pa) detects the resistance of a silicon diaphragm, and thus the space air pressure is then measured. Stainless steel, rubber, etc., can also be used as the material of the diaphragms. A space air pressure sensor should be located in an open area of the conditioned space where the air velocity is less than 40 fpm (0.2 m/s) and where its reading is not affected by the opening of the doors. The reference pressure pickup is best located at the rooftop, at a level 10 ft (3 m) above the building to avoid the influence of wind.

Flow Sensors

Flow sensors usually sense the rate of air flow and water flow in cfm (L/s) for air and gpm (L/s) for water. For airflow sensors, the average velocity pressure P_v in WC (Pa), is often sensed and

measured. The average air velocity v_a , fpm, is then calculated as

$$v_a = 4005K\sqrt{p_v} \quad (5.7)$$

where K is the flow coefficient, which depends on the type of pitot-tube array used and the dimension of the round or rectangular duct. After that, the volume flow rate can easily be determined as the product $v_a A$. Here, A represents the cross-sectional area of the air passage, perpendicular to the airflow, in ft^2 (m^2). Various forms of pitot-tube array have been developed and tested to determine the average velocity pressure of a rectangular or circular duct section by measuring the difference between the total and static pressures of the airstream. A typical pitot-tube flow-measuring station with flow straighteners to provide more even airflow is shown in Fig. 5.11.

Electronic air velocity sensors such as hot-wire anemometers and thermistors have also been widely adopted to measure airflow, especially for variable-air-volume (VAV) boxes. Thermistors are cheaper than hot-wire anemometers. Heated thermistors need periodic calibration.

Water flow sensors of the differential-pressure type, such as orifice plates, flowing nozzles, and pitot tubes, have only a limited measurement range. Turbine or magnet-type flowmeters can apply to a wider range, but they are more expensive and also need periodic calibration.

Carbon Dioxide and Air Quality Sensors

A CO_2 sensor detects and indicates the amount of carbon dioxide (CO_2) contained inside the air, which is a reliable indication of the body odor released by the occupants of the conditioned space. Because of a certain relationship that exists between the CO_2 contained in the outdoor air and the CO_2 released by the occupants, the concentration of CO_2 in space air may sometimes be used as a rough indication of amount of outdoor ventilation air supply to the conditioned space under the specific conditions.

The sensing process used in CO_2 sensors includes potentiometric and amperometric electrochemical cells, an infrared detector, etc. A CO_2 sensor usually has an operating range of 0 to 3000 ppm. For measurements within an accuracy of ± 100 ppm, recalibration may be required on the order of once per year.

An air quality sensor, also called a volatile organic compound (VOC) or a mixed-gas sensor, is used to monitor and detect the relative concentrations of VOC or mixed gas, or total concentration including acetone, ammonia, CO , CO_2 , SO_2 , chlorine, formaldehyde, CFC-11, CFC-12, etc. The concentration of the contaminant is often expressed in tested and commissioned units, such as 0 to 5 or 0 to 10 units.

The sensing element has a tin dioxide surface which is heated to a temperature above 130°F (54°C). The change of the conductivity and thus its resistance are then amplified and fed to the controller in terms of 0 to 10 V dc in proportion to the contamination. Air quality sensors are less expensive than CO_2 sensors and need less maintenance. The drift of an air quality sensor is unpredictable.

Both CO_2 sensors and air quality sensors are now used for demand-based ventilation control to provide the required amount of outdoor air for acceptable indoor air quality.

Occupancy Sensors

An occupancy sensor detects whether a room is occupied by occupant(s). As a result, the HVAC&R and lighting in this room can be turned off when the room is not occupied, to save energy. There are two types of occupancy sensors: ultrasonic and infrared. An ultrasonic occupancy sensor sends out a rather lower level of ultrasonic wave and detects movement of the occupant when there are changes in receiving patterns. Ultrasonic sensors sweep over the area that surrounds the sensor. False signals may be received due to the motion of the papers or air movement from a diffuser. The sensitivity of the ultrasonic sensor should be adjusted to avoid these problems.

An infrared occupancy sensor senses the movement of the occupant(s) as the sensor receives heat from the occupant when that person is moving. Infrared sensors need to see the occupant. As long as there is no obstruction between the sensor and the occupant in the room, the sensing process of an infrared sensor is effective. Occupancy sensors are often mounted on the ceiling or at a high level on a wall. They are used in the guest rooms of the hotels and motels and other commercial buildings. Ultrasonic occupancy sensors show a wider acceptance than infrared occupancy sensors.

Wireless Zone Sensors and Intelligent Network Sensors

In an HVAC&R system, a wireless zone sensor is an indoor spread spectrum radio-frequency transmitter which sends room temperature and other status information to a local receiver located no more than 1000 ft (300 m) away. A translator converts these data and sends them to a variable-air-volume DDC unit controller through a wired communications link. This type of new technology was developed recently for the sake of providing flexibility for the rearrangements of the office layout.

Kovacs (1996) noted that intelligent network sensors "... can linearize the sensor signal, accept an offset adjustment through the network, may have alarm- or decision-making algorithms on board, and include self-test diagnostics to continuously validate performance." An intelligent network sensor may have a sensing element, an analog/digital transducer, a neutron chip—an onboard microprocessor, and a transceiver. Dew-point temperature, enthalpy, wet-bulb temperature, etc. can be provided as a combined variable. An intelligent network sensor will help to drive the system structure towards becoming a more cost-effective distributed control system in the future, a system that has more locally processed signal and data.

Transducers or Transmitters

A transducer is a device that converts energy from one form to another or amplifies an input or output signal. In HVAC&R control systems, a transducer may be used to convert an electric signal to a pneumatic signal (E/P transducer), e.g., a pneumatic proportional relay that varies its branch air pressure from 3 to 15 psig (20 to 103 kPa.g) in direct proportion to changes in the electrical input from 2 to 10 V. Also E/P transducers are used between microprocessor-based or electronic controllers and pneumatic actuators. However, a pneumatic signal can be converted to an electric signal in a P/E transducer. For example, a P/E relay closes a contact when the air pressure falls and opens the contact when the air pressure rises above a predetermined value. A transmitter is used to transmit a signal, pneumatic or electric; through air, water, or other fluids. A sensor is also a transmitter. However, the difference between them is that a sensor only senses the signal of the controlled variable and transmits it.

5.5 CONTROLLERS

A controller receives input from the sensor, compares with the set point or implementation based on its stored computer software, sends an output to or modulates the control device for maintaining a desirable indoor environment. A thermostat is a combination of a temperature sensor and a temperature controller, whereas a humidistat is a combination of a humidity sensor and a humidity controller.

Direct-Acting or Reverse-Acting

A direct-acting controller increases its output signal upon an increase in the sensed controlled variable, and it decreases its output signal upon a decrease in the sensed controlled variable. Conversely,

a reverse-acting controller decreases its output signal upon an increase in the sensed controlled variable and increases its output signal upon a decrease in the controlled variable.

Normally Closed or Normally Open

A controlled device, a valve or a damper, that is said to be in the normally closed (NC) position indicates that when the input signal to the controller falls to zero or below a critical value, the control device will be in the closed position. The closing of the valve or damper is most likely due to the action of a spring or a supplementary power supply. A valve or a damper that is said to be normally open (NO) indicates that when the input signal to the controller falls to zero or below a critical value, then the valve, damper, or associated process plant will be in the open position.

Pneumatic Controllers

In a pneumatic controller, the basic mechanism used to control the air pressure in the branch line supplied to the actuators is a nozzle-flapper assembly plus a restrictor. Figure 5.12 shows a pneumatic controller with such a mechanism. Compressed air supplied from the main line flows through the restrictor and discharges at the opening between the nozzle and the flapper, which has a spring pulling it downward. The nozzle and restrictor are sized in such a manner that when the flapper moves away from the nozzle, all the air escapes from the nozzle and the branch line pressure is zero. When the flapper covers the nozzle or if there is no airflow in the branch line, the pressure of the branch line is equal to that in the main line.

If a sensor, such as a bimetal sensor, moves the flapper upward according to the magnitude of the controlled variable sensed by the sensor, then the input signal from the sensor determines the opening between the nozzle and flapper and hence the compressed air pressure in the branch line.

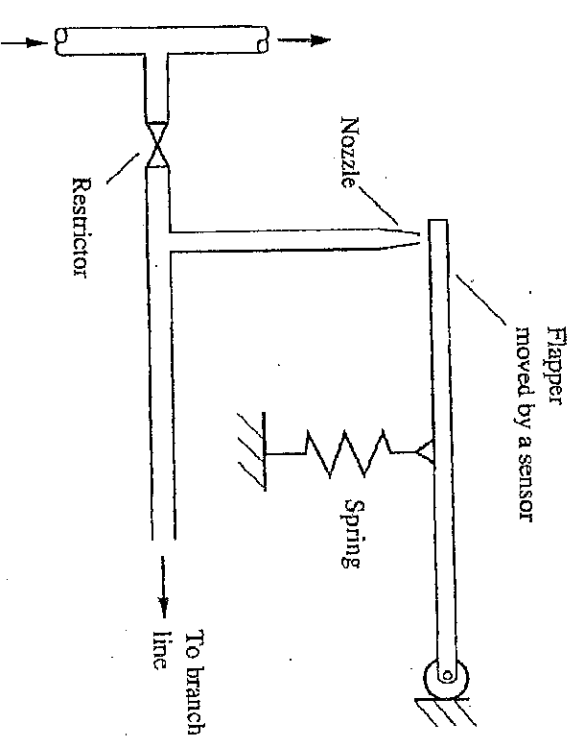


FIGURE 5.12 A pneumatic controller with a nozzle-flapper assembly and a restrictor.

For a direct-acting pneumatic temperature controller used to control the space temperature during summer, the branch line pressure may change from 9 to 13 psig (62 to 89 kPag) when the space temperature increases from 73 to 77°F (22.8 to 25°C). This nozzle-flapper assembly pneumatic controller operates in proportional control mode. Many other more complicated pneumatic controllers have been developed to perform other control modes and additional functions.

Electric and Electronic Controllers

An electric controller uses switches, relays, and a bridge circuit formed by potentiometers to position the actuators in on-off, floating, and proportional control modes according to the input signal from the sensor and the predetermined set point. An electronic controller can provide far more functions than electric controllers can. It may receive input signals from both the main sensor and the compensation sensor with amplification and combination.

In the control circuit, an electronic controller basically provides proportional or proportional-integral control modes. The output signal from the controller can be used to position an actuator or to provide the sequencing of actuators, or to change to two-position, floating, or even PID control modes in conjunction with additional circuits.

Direct Digital Controllers

A direct digital controller has a microprocessor to implement computer programs to provide various control functions. In DDC units, there are analog-to-digital (A/D) and digital-to-analog (D/A) converters to convert analog input to digital signals for processing, or to convert digital signals to analog for actuators, if necessary. DDC units are stand-alone and microprocessor-based controllers. *Stand-alone* means that the controller has sufficient capacity to execute the assigned control functions alone. Today, there are mainly two types of DDC units: system controllers and unit controllers.

System Controllers. A system controller, also called a stand-alone panel (SAP), as shown in Fig. 5.13, has the ability to coordinate communications between system controllers, between the system controller and the personal computer (PC) in the workstation, and between the system controller and the supported unit controllers. It also has the ability to provide and execute the control programs for functional control, and to store user databases and trend log values. A system controller can support 50 to 200 unit controllers on separate unit controller trunk(s).

Unit Controllers. A unit controller, also called a terminal controller, is shown in Fig. 5.14. It usually has limited capacity to execute factory-loaded computer programs and to provide functional control for a terminal or a piece of HVAC&R equipment. Unit controllers are often connected in a separate network and supported by a system controller. The new-generation unit controllers have greater memory to handle complicated control programs, and they provide time and calendar scheduling, data storage, and other functions, such as limited programming.

Hardware. Many system controllers have only a single printed-circuit board. Single-board configuration often offers lower first cost. Its disadvantage is that any component failure requires the replacement of the complete board. Another approach is that the controller is made from various modules. The modular approach isolates the component failures and plugs on the single-module replacement quickly and inexpensively.

Memory. The types of memory included in DDC units are as follows:

- Read-only memory (ROM), which stores the software provided by the manufacturer and should not be modified by the user.

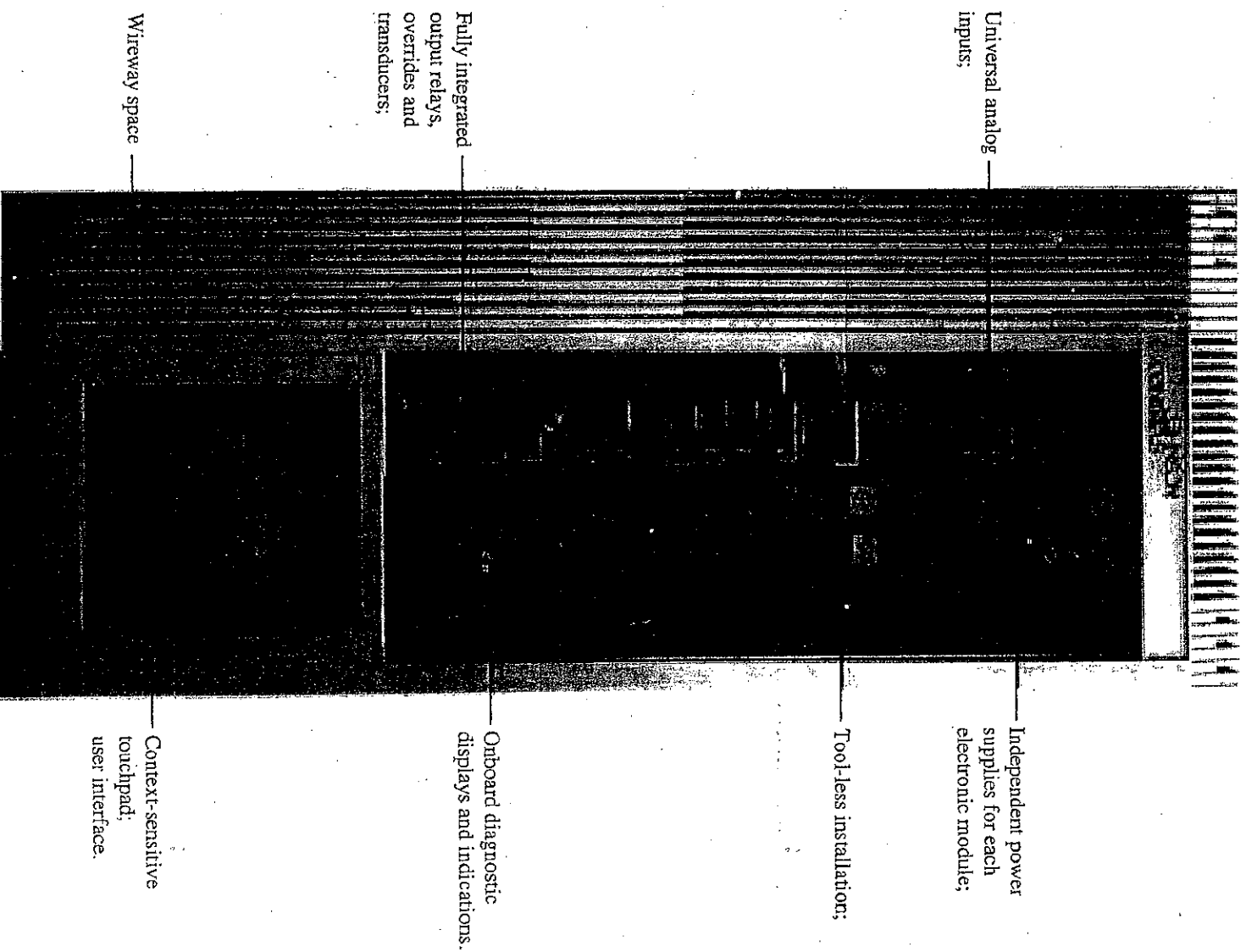


FIGURE 5.13 A typical system controller. (Source: Johnson Controls. Reprinted by permission.)

- Random-access memory (RAM), which stores custom control software developed during installation or prepared by the user. This type of memory is volatile (i.e., it can be read from and written to) and requires battery backup.
- Electric erasable programmable read-only memory (EEPROM), which stores custom control software and is volatile. The advantage of EEPROM over RAM is that EEPROM does not need

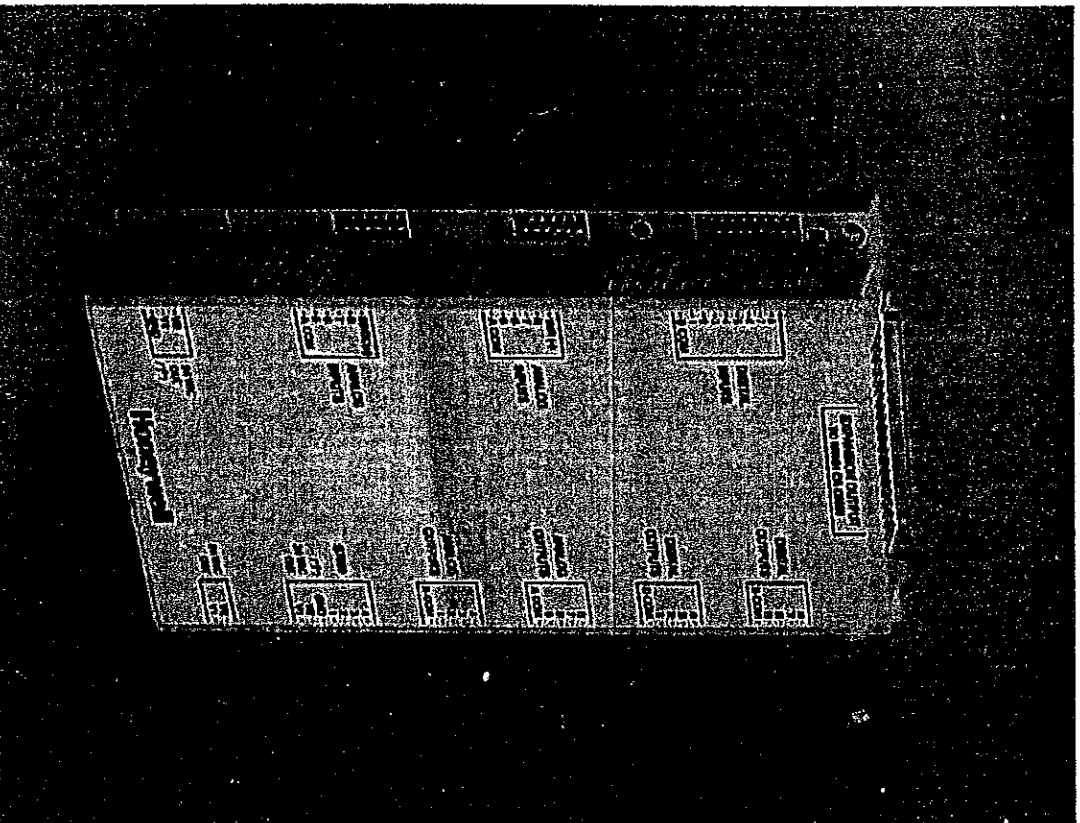


FIGURE 5.14 A typical unit controller. (Source: Honeywell Inc. Reprinted by permission.)

battery backup. However, EEPROM cannot be used for as many writing, erasing, and rewriting cycles as RAM.

- Flash erasable programmable read-only memory (flash EPROM), which is a kind of new memory technology and allows the stored control software to remain untouched indefinitely without power.

System controllers often use a 16-bit microprocessor. A typical system controller has the following memories:

RAM:	256 Kbytes
ROM:	128 Kbytes EEPROM
	512 Kbytes flash EPROM

Input/Output (I/O). There are four kinds of I/O: analog input (AI); digital or binary input (BI); analog output (AO); and pulsed or binary output (BO). Conventionally, a sensor input, a controller output, or a control value, is referred to as a *point* or *object*. Current system controllers allow their

input and output connections to be configured with great flexibility. Each input or output point can be either analog or digital. Typical analog inputs (electric signal) are 0 to 10 V dc or 4 to 20 mA. A system controller often has a total of 20 to 50 I/O points. Some system controllers can be extended to 100 points if necessary. A typical system controller has the following I/O points capacity:

Analog/digital inputs	18
Universal analog/digital input/outputs	6
Digital outputs	12
Totalizer inputs, i.e., pulsed inputs	4

A unit controller or a terminal controller usually has 4 to 20 I/O points. The I/O points in a typical unit controller may take the following forms:

Analog inputs	0 to 10 V dc
Digital inputs	Switch, relay, and transistor
Microbridge sensor	0 to 3 in. WC pressure differential (0 to 750 Pa)
Analog outputs	0 to 10 V dc, 4 to 20 mA
Digital outputs	30 V ac
Pneumatic	3 to 15 psig (20.6 to 103 kPag)
Triac	On/off output for electric heater, fan motor, etc.

5.6 WATER CONTROL VALVES AND VALVE ACTUATORS

Water valves are used to regulate or stop water flow in a pipe either manually or by means of automatic control systems. Water control valves adopted in water systems can modulate water flow rates by means of automatic control systems.

Valve Actuators

An actuator, sometimes called an operator, is a device which receives an electric or pneumatic analog control signal from the controller, either directly or through a digital-to-analog converter. It then closes or opens a valve or damper, modulating the associated process plant, and causes the controlled variable to change toward its set point. Valve actuators are used to position control valves. They are mainly of the following types:

Solenoid Actuators. These use a magnetic coil to move a movable plunger connected with the valve stem. Most solenoid valve actuators operate at two positions (on and off). They are used mainly for small valves.

Electric Actuators. These move the valve stem by means of a gear train and linkage. Different electric motor valve actuators can be classified according to the control mode they use:

1. *On/off mode.* For this type of actuator, the motor moves the valve in one direction, and when the electric circuit breaks, the spring returns the valve stem to the top position (either open or closed position depending on whether it is a normally open or closed valve).
2. *Modulating mode.* The motor can rotate in both directions, with spring return when the electric circuit breaks.
3. *Modulating mode with supplementary power supply.* The motor rotates in two directions and without a spring-return arrangement. When the power is cut off, a bypass signal is usually sent

to the electric motor to drive the valve to its open or closed position, depending on whether it is a normally open or closed valve.

It may take minutes to fully open a large valve using an electric motor valve actuator. Modern electronic actuators use solid-state control boards to determine the speed, the action, and other functions to meet more demanding requirements.

Pneumatic Actuators. A pneumatic valve actuator consists of an actuator chamber whose bottom is made of a flexible diaphragm or bellows connected with the valve stem. When the air pressure in the actuator chamber increases, the downward force overcomes the spring compression and pushes the diaphragm downward, closing the valve. As the air pressure in the actuator chamber decreases, the spring compresses the diaphragm, moving the valve stem and valve upward. A pneumatic valve actuator is powerful, simple, and fast to respond.

Because of the increasing popularity of the DDC systems, there is an increasing demand for electric actuators that can be interfaced with a DDC system.

Types of Control Valves

Water control valves consist mainly of a valve body, one or two valve disks or plugs, one or two valve seats, a valve stem, and a seal packing. Based on their structure, water control valves can be classified into the following types:

1. *Single-seated.* A single-seated valve has only a single valve disk and seal, as shown in Fig. 5.15b and c. It is usually used for water systems that need a tight shutoff.
2. *Double-seated.* A double-seated valve has two valve disks connected to the same valve stem and is designed so that the fluid pressure exerted on the valve disks is always balanced. Consequently, less force is required for the operation of a double-seated valve, as shown in Fig. 5.15a.
3. *Butterfly.* A butterfly valve consists of a cylindrical body, a shaft, and a disk that rotates on an axis, as shown in Fig. 5.15d. When the valve closes, it seats against a ring inside the body. A butterfly valve exhibits low flow resistance when it is fully opened. It is compact and is usually used in large water pipes.

According to the pattern of the water flow, water control valves can again be classified as two-way valves or three-way valves. A two-way valve has one inlet port and one outlet port. Water flows straight through the two-way valve along a single passage, as shown in Fig. 5.15a.

In a three-way valve, there are three ports: two inlet ports and one common outlet port for a three-way mixing valve, as shown in Fig. 5.15b, and one common inlet port and two outlet ports for a three-way diverting valve, as shown in Fig. 5.15c. In a three-way mixing valve, the main water stream flows through the coil or boiler, and the bypass stream mixes with the main stream in the common mixing outlet port. In a three-way diverting valve, the supply water stream divides into two streams in the common inlet port. The main water stream flows through the coil, and the bypass stream mixes with the main water stream after the coil. A three-way mixing valve is always located downstream of the coil. But a diverting valve is always located upstream of the coil.

A diverting valve should never be used as a mixing valve. The unbalance pressure difference between the two inlet ports and the outlet port at a closed position may cause disk bouncing and valve wear when the valve disk travels between the two extremes.

Valve Characteristics and Ratings

The different types of control valves and the characteristics that are important during the selection of these valves are as follows:

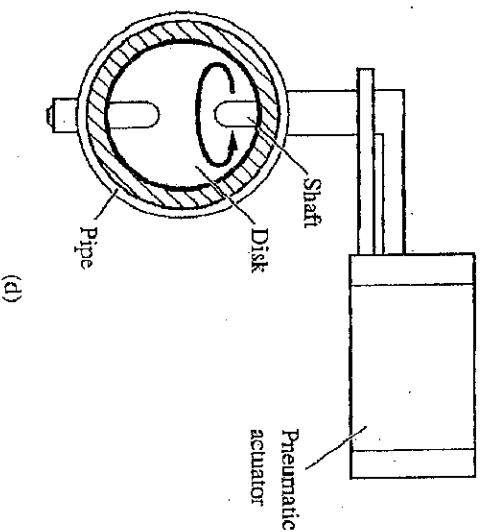
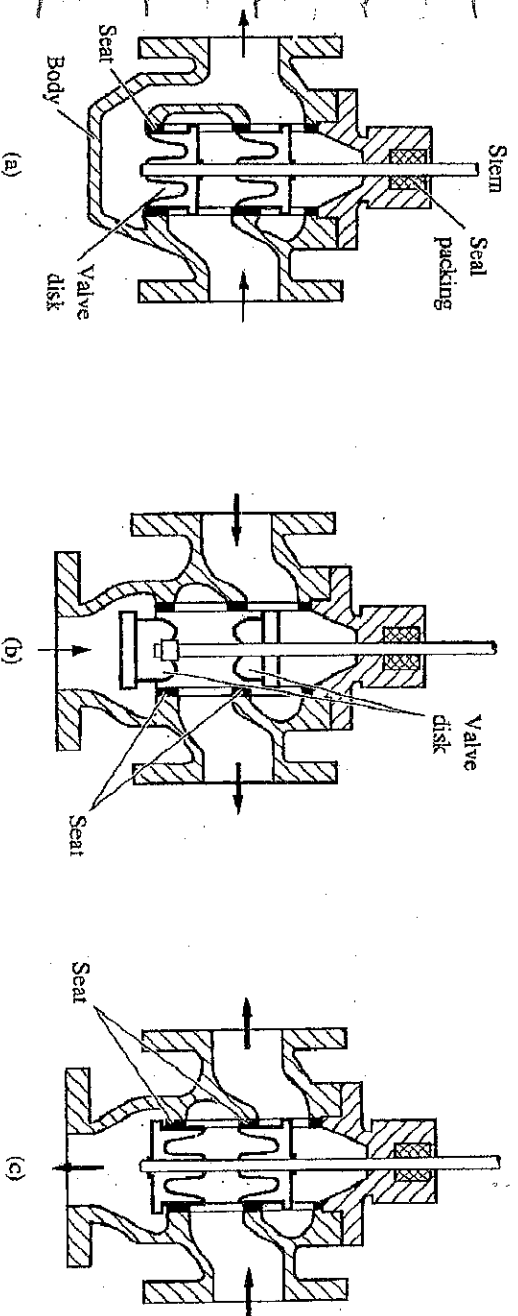
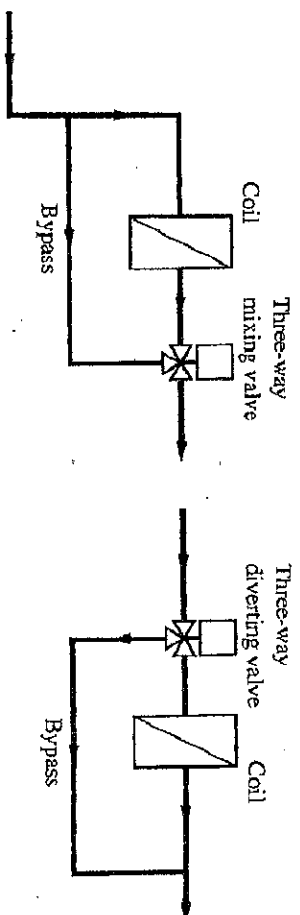


FIGURE 5.15 Various types of control valves: (a) Double-seated two-way valve; (b) single-seated three-way mixing valve; (c) single-seated three-way diverting valve; (d) butterfly valve.

1. *Equal-percentage valve.* This control valve changes the water flow rate by a certain percentage for that same percentage of lift in the valve stem when the upstream versus downstream water pressure difference across the valve (its pressure drop) is constant.
2. *Linear valve.* This control valve shows a directly proportional relationship between the flow rate and the lifting of the valve stem for a constant pressure drop.

3. *Quick-opening valve.* This control valve gives the maximum possible flow rate when the valve disk or plug is just lifted from its seat.

Rangeability is defined as the ratio of the maximum flow rate to the minimum flow rate under control. An equal-percentage valve may have a very good rangeability of 50:1. A linear valve may have a rangeability of 30:1.

The flow characteristics of equal-percentage, linear, and quick-open valves are shown in Fig. 5.16.

The following control valve ratings should be considered during the selection and sizing of a valve:

1. *Body rating.* The nominal body rating of the valve is the theoretical rating of the valve body only, in psig. The actual body rating is the permissible safe water pressure for the valve body, in psig (kPa·g), at a specific water temperature.
2. *Close-off rating.* That is the maximum pressure difference between the inlet and outlet ports that a valve can withstand without leakage when the valve is fully closed, in psi (kPa).
3. *Maximum pressure and temperature.* These are the maximum pressure and temperature of water that the whole valve, including body, disk, seat, packing, etc., can withstand.

Valve Selection

Proper selection of water control valves depends on water system performance, load variations, pipe size, control modes, etc. Today, the use of scaling factors on analog outputs in a DDC system permits a nonlinear device to provide an output of linear response. However, select a control valve having a linear relationship between a change in the controlled variable and the amount of travel of the valve stem, or a linear system control characteristic over the operating range is still desirable when it is cost-effective. Hence, a linear valve is often used for the water system for which the controlled variable has

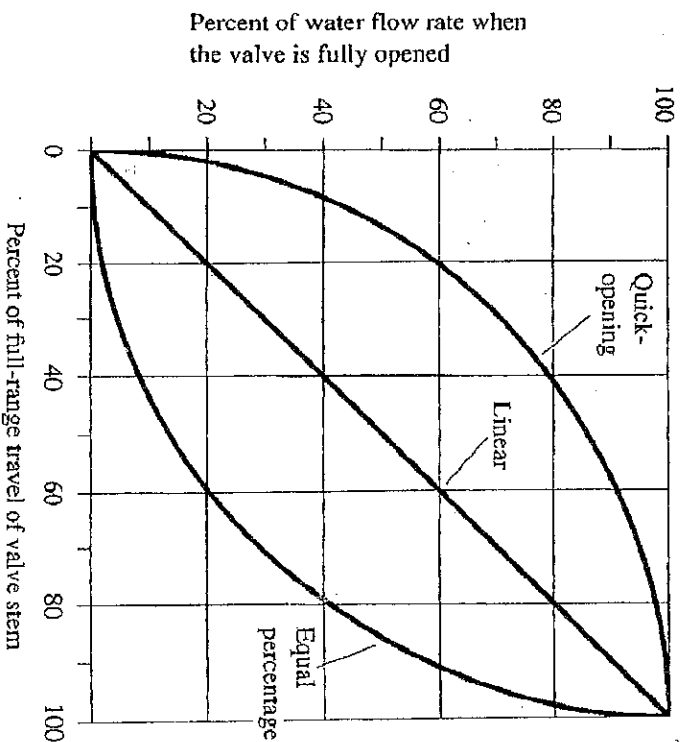


FIGURE 5.16 Typical flow characteristics of various types of control valve.

a linear relationship with the water flow or valve opening; or for applications that do not have wide load variations.

When a control valve is used to modulate the water flow rate of a hot or chilled water coil, a large reduction in the flow rate causes only a small reduction in the heating or cooling output of the coil. Given such circumstances, the nonlinear behavior of an equal-percentage valve combined with the nonlinear output performance of a hot or chilled water coil will provide more linear system behavior.

When three-way valves are used, the water flow rate before or after the common port is approximately constant, no matter how wide the openings of the various ports in the three-way valves. As such, three-way valves are used in constant or approximately constant water flow rate systems, even the coil load changes.

As a two-way valve closes, the flow rate of the water system decreases and its pressure drop across the two-way valve increases. A two-way valve is thus used for water systems that have variable volume flow during a variation in system load, as shown in Fig. 5.17.

Valve Sizing

The size of a control valve affects the controllability of a water system. If a control valve is oversized, then the smallest increment possible may overshoot the controlled variable. An undersized control valve needs great pumping power. The size of a control valve is also closely related to its

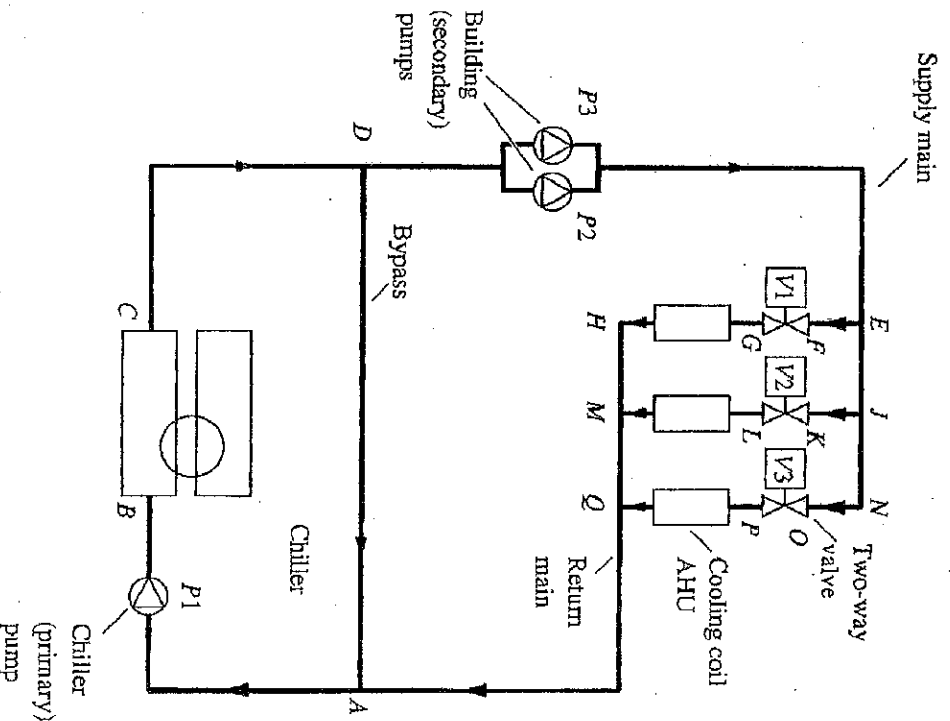


FIGURE 5.17 A typical chilled water system.

design water flow rate V , gpm (L/s), and the pressure drop across the valve ΔP_{vw} psi (kPa), when the control valve is fully opened. Their relationship can be expressed as

$$V = C_v \sqrt{\Delta P_{vw}} \quad (5.8)$$

$$C_v = \frac{V}{\sqrt{\Delta P_{vw}}}$$

where C_v is the flow coefficient for a flow rate of 1 gpm at a pressure drop of 1 psi. The flow coefficient of control valves can be found in manufacturers' catalogs.

For a water system that has several cooling coils between supply and return mains, as shown in Fig. 5.17, modulating the water flow is effective only when the opening and closing of the control valve $V1$ in piping section EH has a significant effect on the change in the pressure drop ΔP_{EH} across piping section EH between the supply and return mains. As control valve $V1$ opens wider and ΔP_{EH} decreases, if there is no concurrent change in flow resistance of piping sections JM and NQ , then a greater water flow will pass through piping section EH . If ΔP_{EH} increases, less water will flow through EH . If ΔP_{EH} , ΔP_{JM} , and ΔP_{NQ} all increase, based on the characteristic curve of the centrifugal pump, the water flow rate through pumps $P2$ and $P3$ will decrease for an increase in the system head. If ΔP_{EH} , ΔP_{JM} , and ΔP_{NQ} all drop, the water flow rate through pump $P2$ or $P3$ will increase.

Because of the use of DDC systems with PI or PID control modes and the variable-speed pumping systems, as well as the results of the previous analysis the following hold:

1. The size of a control valve should be determined according to the flow coefficient calculated from Eq. (5.8) and listed in the manufacturer's catalog.
2. The assumed pressure drop ΔP_{vw} across the control valve should be appropriate. It is affected by the type of control valve used. It is also a compromise between a higher ΔP_{vw} value to provide desirable controllability and a lower ΔP_{vw} value to save energy.
3. For a water system using variable-speed pumping and DDC systems with PI or PID control modes, a pressure drop across an equal-percentage control valve $\Delta P_{vw} = 5$ to 10 ft WC (15 to 30 kPa) and a rangeability of 30 : 1 or greater is recommended for energy saving.

Example 5.1. A water system using variable-speed pumping with DDC supplies chilled water to the cooling coils of three air-handling units (AHUs), as shown in Fig. 5.17. If the chilled water flowing through two-way valve $V1$ is at 100 gpm (6.31 L/s), select and size control valve $V1$.

Solution.

1. To achieve a nearly linear relationship between the coil load and the travel of the valve stem, an equal-percentage valve is selected for valve $V1$.
2. For a water system using variable-speed pumping with DDC, it is assumed that the pressure drop across the control valve $\Delta P_{vw} = 5$ ft WC, or $5 \times 0.433 = 2.165$ psi (15 kPa). From Eq. (5.8), the flow coefficient is

$$C_v = \frac{V}{\sqrt{\Delta P_{vw}}} = \frac{100}{\sqrt{2.165}} = 68$$

From one of the manufacturers' catalog, for $C_v = 68$, the size of equal-percentage control valve $V1$ is 3 in. (76 mm).

3. From the friction chart of water in steel pipes, for a chilled water flow rate of 100 gpm (6.31 L/s) and a piping head loss of 2.5 ft/100 ft (2.5 m/100 m) of length, the diameter of the branch piping section EH is also 3 in. (76 mm). The size of the control valve and the diameter of the branch pipe are the same.

5.7 DAMPERS AND DAMPER ACTUATORS

A damper is a device that controls the airflow in an air system or ventilating system by changing the angle of the blades and therefore the area of its flow passage. In HVAC&R systems, dampers can be divided into volume control dampers and fire dampers. Fire dampers are covered in a later section. In this section, only volume control dampers are discussed.

Types of Volume Control Dampers

Volume control dampers can be classified as single-blade dampers or multiblade dampers according to their construction. Various types of volume control dampers are shown in Fig. 5.18.

Butterfly Dampers. A butterfly damper is a single-blade damper. A butterfly damper is made from either a rectangular sheet mounted inside a rectangular duct or a round disk placed in a round duct, as shown in Fig. 5.18a. It rotates about an axle and is able to modulate the air volume flow rate of the duct system by varying the size of the opening of the passage for air flow.

Gate Dampers. A gate damper is a single-blade damper. It also may be rectangular or round. It slides in and out of a slot in order to shut off or open up a flow passage, as shown in Fig. 5.18b. Gate dampers are mainly used in industrial exhaust systems with high static pressure.

Split Dampers. A split damper is also a single-blade damper. It is a piece of movable sheet metal that is usually installed at the Y connection of a rectangular duct system, as shown in Fig. 5.18c.

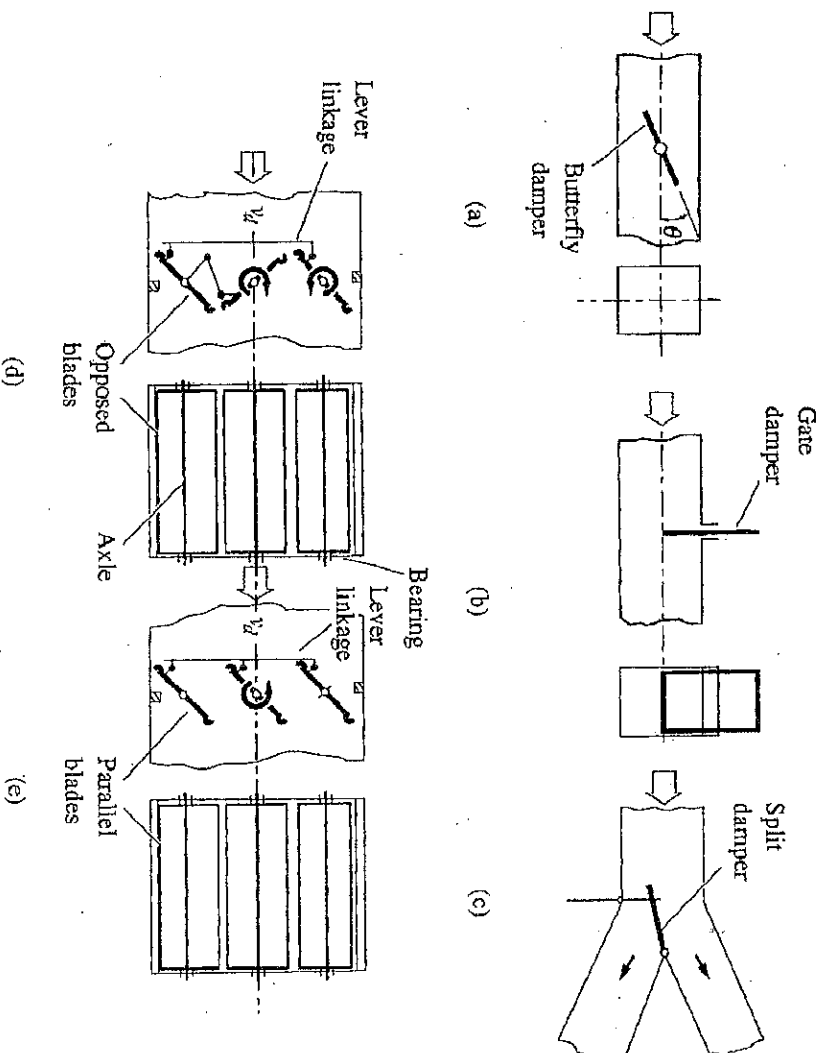


FIGURE 5.18 Various types of volume control dampers: (a) Butterfly damper; (b) gate damper; (c) split damper; (d) opposed-blade damper; (e) parallel-blade damper.

The movement of the split damper from one end to the other modulates the volume of air flowing into the two legs or branches. A split damper is usually modulated only during air balancing after installation or during periodic air balancing.

Opposed-Blade Dampers. An opposed-blade damper is a type of multiblade damper that is often rectangular, as shown in Fig. 5.18d. It is usually used for a flow passage of large cross-sectional area. The damper blades may be made of galvanized steel, aluminum alloy, or stainless-steel sheets, usually not exceeding 10 in. (25.4 cm) in width. Rubber or spring seals can be provided at the fully closed position to control the air leakage rating, which often does not exceed 6 cfm/ft^2 ($30 \text{ L/s}\cdot\text{m}^2$) at a pressure drop across the damper of 4 in. WC (1000 Pa). The bearing used for supporting the blade axle should be made of a corrosion-resistant material such as copper alloy or nylon. Teflon-coated bearings may also be used to ensure smooth operation of the damper. Lever linkages are used to open and close the damper blades. The characteristics of the opposed-blade dampers are covered later in this section.

The maximum static pressure drop across closed opposed-blade dampers is 6 in. WC (1500 Pa) for a 36-in.- (914-mm-) long damper (the length of the damper blade) and 4 in. WC for a 48-in.- (1219-mm-) long damper.

Parallel-Blade Dampers. A parallel-blade damper is also a type of multiblade damper used mainly for large cross-sectional areas, as shown in Fig. 5.18e. The blade material and the requirement for the seals and bearings are the same as those for opposed-blade dampers.

Damper Actuators (Motors)

Damper actuators, also called damper motors, are used to position dampers according to a signal from the controller. As with valve actuators, damper motors can be classified as either electric or pneumatic.

Electric Damper Motors. These either are driven by electric motors in reversible directions or are unidirectional and spring-returned. A reversible electric motor is used often for more precise control. It has two sets of motor windings. When one set is energized, the motor's shaft turns in a clockwise direction; and when the other set is energized, the motor's shaft turns in a counterclockwise direction. If neither motor winding is energized, the shaft remains in its current position. Such an electric motor can provide the simplest floating control mode, as well as other modes if required.

Pneumatic Damper Motors. Their construction is similar to that of pneumatic valve actuators, but the stroke of a pneumatic damper motor is longer. They also have lever linkages and crank arms to open and close the dampers.

Volume Flow Control between Various Airflow Paths

For air conditioning control systems, most of the dampers are often installed in parallel connected airflow paths to control their flow volume, as shown in Fig. 5.19. The types of airflow volume control are as follows:

Mixed-Air Control. In Fig. 5.19a, there are two parallel airflow paths: the recirculating path *w_{rn}* in which a recirculating air damper is installed and the exhaust and intake path *w_{om}*, in which exhaust and outdoor air dampers are installed. The outdoor air and the recirculating air are mixed together before entering the coil. Both the outdoor damper and the recirculating damper located just before the mixing box (mixed plenum) are often called mixing dampers. The openings of the outdoor and recirculating dampers can be arranged in a certain relationship to each other. When the outdoor damper is at minimum opening for minimum outdoor air ventilation, the recirculating

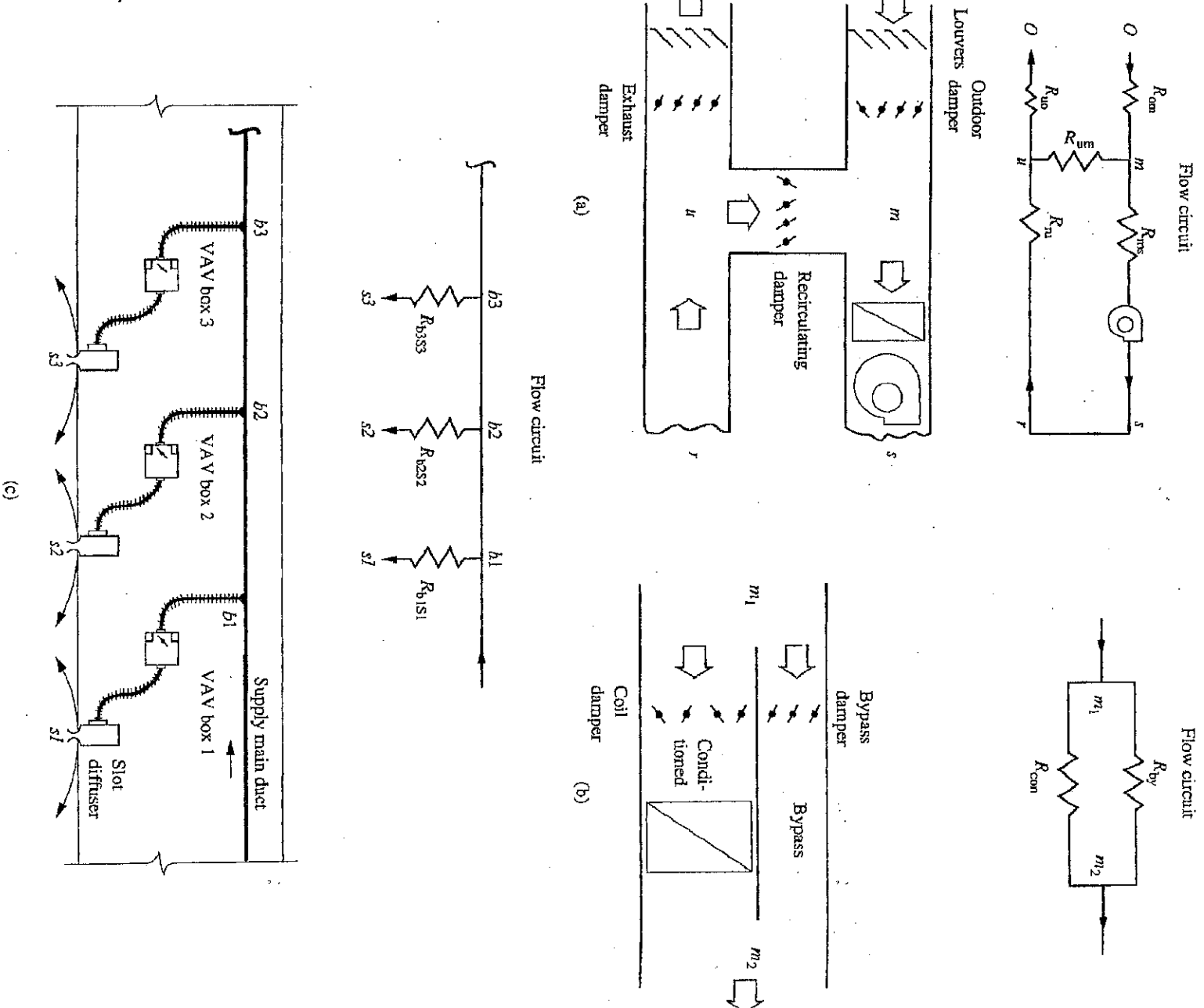


FIGURE 5.19 Airflow paths: (a) mixed-air control, (b) bypass control, and (c) branch flow control.

damper is then fully opened. If the outdoor damper is fully opened for free cooling, the recirculating damper is closed.

Bypass Control. In the flow circuit for bypass control, as shown in Fig. 5.19b, the entering air at the common junction m_1 is divided into two parallel airflow paths: the bypass path, in which a bypass damper is installed, and the conditioned path, in which the coil face damper is installed in series with the coil, or the washer damper with the air washer. The bypass and the conditioned airstreams are then mixed together at the common junction m_2 . The face and bypass dampers can also be arranged in a certain relationship to each other.

Branch Flow Control. In a supply main duct that has many branch take offs, as shown in Fig. 5.19c, there are many parallel airflow path combinations: paths $b1s1$ and $b1b2s2$, $b2s2$ and $b2b3s3$, etc. In each branch flow path, there is a damper in the VAV box, and points $s1$, $s2$, $s3$, etc., are the status points of the supply air.

Parallel airflow paths such as those shown in Fig. 5.19 have the following characteristics:

1. The total pressure losses of the two airflow paths that connect the same endpoints are always equal; for example, $\Delta p_{um} = \Delta p_{nom}$, $\Delta p_{m1\ bypass2} = \Delta p_{m1\ comm2}$, etc.
2. The relationship between total pressure loss Δp , in WC (Pa); flow resistance R , in WC/(cfm)² (Pa·s²/m⁶); and volume flow rate V , cfm (m³/s), can be expressed as

$$\Delta p = RV^2 \quad (5.9)$$

Flow resistance is covered in greater detail in Chap. 10.

3. If the total pressure loss Δp remains constant and the flow resistance R , of one parallel path increases, from Eq. (5.9), the airflow through this path V must be reduced. The airflow in other parallel paths remains the same.
4. The total pressure loss of an airflow path between two common junctions Δp determines the volume flow rate of air passing through that path and can be calculated from Eq. (5.9) as

$$V = \sqrt{\frac{\Delta p}{R}}$$

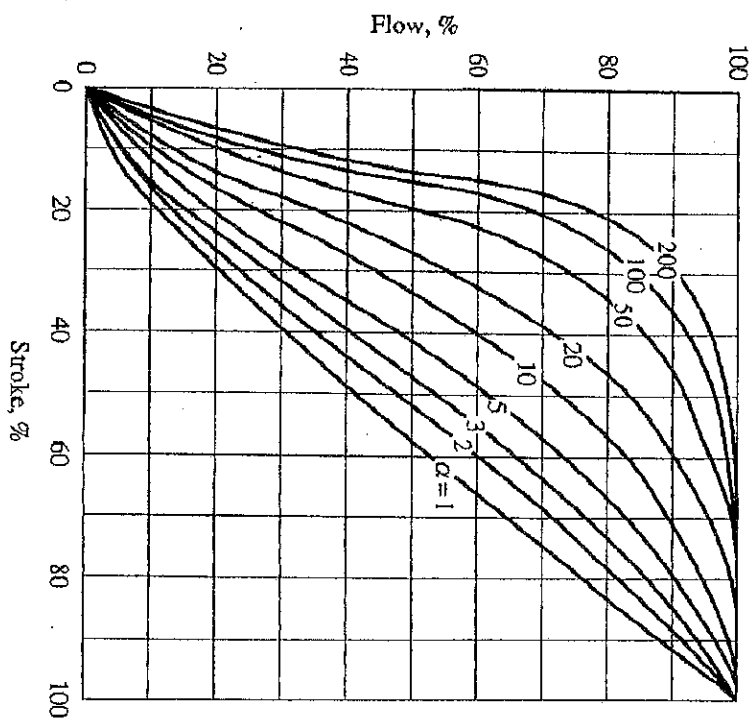
5. When the flow resistances in most of the branches increase because of the closing of the dampers to a small opening in their VAV boxes, the flow resistance of the supply duct system R_{sys} and the system total pressure loss Δp_{sys} both tend to increase, and thus the total air volume flow of the supply duct system V_{sys} will reduce accordingly.

Flow Characteristics of Opposed- and Parallel-Blade Dampers

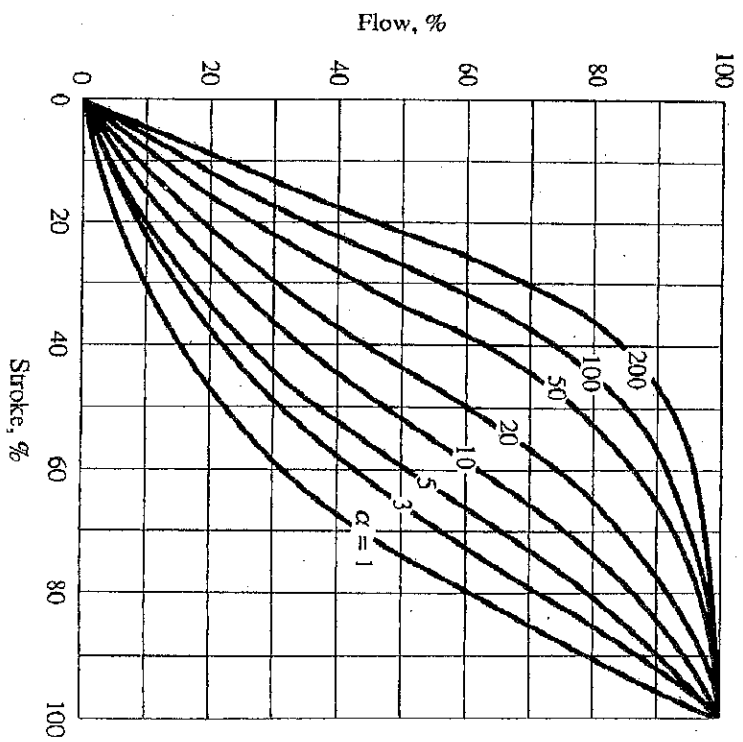
A parallel-blade or an opposed-blade damper that is installed in a single airflow path to modulate airflow is often called a volume control damper (or throttling damper). For volume control dampers, a linear relationship between the percentage of the damper opening and the percentage of full flow is desirable for better controllability and cost effectiveness. (Full flow is the air volume flow rate when the damper is fully opened at design conditions.) The actual relationship is given by the installed characteristic curves of parallel-blade and opposed-blade dampers shown in Fig. 5.20a and b. For the sake of energy savings, it is also preferable to have a lower pressure drop when air flows through the damper at the fully open condition.

In Fig. 5.20, α is called the *damper characteristic ratio* and may be calculated as

$$\alpha = \frac{\Delta p_{path} - \Delta p_{od}}{\Delta p_{od}} = \frac{\Delta p_{path}}{\Delta p_{od}} - 1 = \frac{\Delta p_{p-od}}{\Delta p_{od}} \quad (5.10)$$



(a)



(b)

FIGURE 5.20 Flow characteristic curves for dampers: (a) parallel-blade and (b) opposed-blade.

where Δp_{path} = total pressure loss of airflow path, in. WC (Pa)

Δp_{od} = total pressure loss of the damper when it is fully opened, in. WC (Pa)

$\Delta p_{\text{p-od}}$ = total pressure loss of air flow path excluding damper, in. WC (Pa)

Damper Selection

Butterfly dampers are usually used in ducts of small cross-sectional area or in places like VAV boxes.

For volume control dampers in a single airflow path, in order to have better controllability, an opposed-blade damper is recommended if many dynamic losses other than the damper itself (such as coil or air washer, heat exchanger, and louvers) exist in the airflow path. If the damper is the primary source of pressure drop in the airflow path, a parallel-blade damper is often used.

For mixing dampers, a parallel-blade damper is recommended for the recirculating damper as the pressure drop across the damper is often the primary source in its airflow path. An opposed-blade damper is recommended for the outdoor damper and exhaust (relief) damper for better controllability. The parallel blades of the recirculating damper should be arranged so that the recirculating airstream will blow toward the outdoor airstream, resulting in a more thorough mixing. Many packaged units also use parallel-blade outdoor dampers for smaller pressure drop and less energy consumption.

For face and bypass dampers, an opposed-blade coil face damper in an airflow path of greater pressure drop and a parallel-blade bypass damper will give better linear system control characteristics.

For two-position control dampers, a parallel-blade damper is always used because of its lower price.

Damper Sizing

Damper sizing should be chosen to provide better controllability (such as a linear relationship between damper opening and airflow), to avoid airflow noise if the damper is located in the ceiling plenum, and to achieve an optimum pressure drop at design flow to save energy.

The face area of the damper A_{dam} , ft^2 (m^2), in most cases is smaller than the duct area A_d , in ft^2 (m^2). Based on Alley (1988) paper, the local loss coefficient C_{dam} of the damper for different setups can be determined from Fig. 5.21. Then the pressure drop across the damper when the damper is fully opened Δp_{od} , in. WC (Pa), can be calculated as

$$\Delta p_{\text{od}} = C_{\text{dam}} \left(\frac{V_{\text{dam}}}{4005} \right)^2 \quad (5.11)$$

$$V_{\text{dam}} = \frac{V}{A_{\text{dam}}} \quad (5.12)$$

where V_{dam} = face velocity of the damper, fpm.

1. The damper is generally sized when the air flowing through the damper is at a maximum. For an outdoor damper, the maximum airflow usually exists when the free cooling air economizer cycle is used. For a recirculating damper, its maximum airflow occurs when the outdoor air damper is at minimum opening position, to provide outdoor air ventilation.

2. The face velocity of dampers V_{dam} is usually 1000 to 3000 fpm (5 to 15 m/s), except that the face velocity of a butterfly damper in a VAV box may drop to only 500 fpm (2.5 m/s) for energy savings and to avoid airflow noise. The ratio A_{dam}/A_d is often between 0.5 and 0.9.

3. The outdoor damper may be either made in a one-piece damper or split into two dampers, a larger and a smaller, to match the needs at free cooling and minimum outdoor ventilation.

4. For a bypass damper, its face area should be far smaller than that of an air washer or than a water heating or cooling coil's face damper. When the air washer or coil's face damper is closed, the area of the bypass damper should provide an airflow that does not exceed the system design airflow.

11.5 CAPACITY AND SAFETY CONTROLS

Capacity Control

Refrigeration systems for comfort air conditioning or industrial applications need capacity control to meet refrigeration load variations during operation. In vapor compression refrigeration systems, control of the refrigerant flowing through the compressor is widely used as the primary refrigeration capacity control. For reciprocating refrigeration systems using DX coils as evaporators, row or intertwined face control of DX coil capacity is often employed in conjunction with compressor capacity control for better performance. In medium-size and large systems, two separate refrigerant circuits are often used so that refrigerant flow in one can be cut off when the refrigeration load drops below 50 percent of design load.

There are four basic methods of controlling the capacity of reciprocating refrigeration systems by reducing the refrigerant flow rate during part-load operation.

On/Off Control. On/off or start/stop control is widely used to control the refrigerant flow and the capacity of the reciprocating compressors in residential air conditioners. For reciprocating refrigeration systems equipped with multiple compressors, as used in supermarkets, on/off control is often the suitable choice. On/off control is simple and inexpensive, but it produces a greater variation in the space air temperature and relative humidity maintained by the refrigeration system than other types of control do. In addition, cyclic operation may lead to a start/stop cyclic loss for the evaporator and condenser, which are located indoors and outdoors separately. Cyclic loss is discussed in the next chapter. On/off control also results in increased wear and tear on compressor components and shortens service life.

Cylinder Unloader. For multiple-cylinder reciprocating compressors, a cylinder unloader is often used for capacity control. Cylinder unloaders reduce the capacity of compressors by bypassing the compressed gas to the suction chamber, blocking the suction or discharge valve, or closing the suction valve late or early. Recently developed cylinder unloaders include a solenoid valve that unloads the cylinder(s) in response to any one of the following:

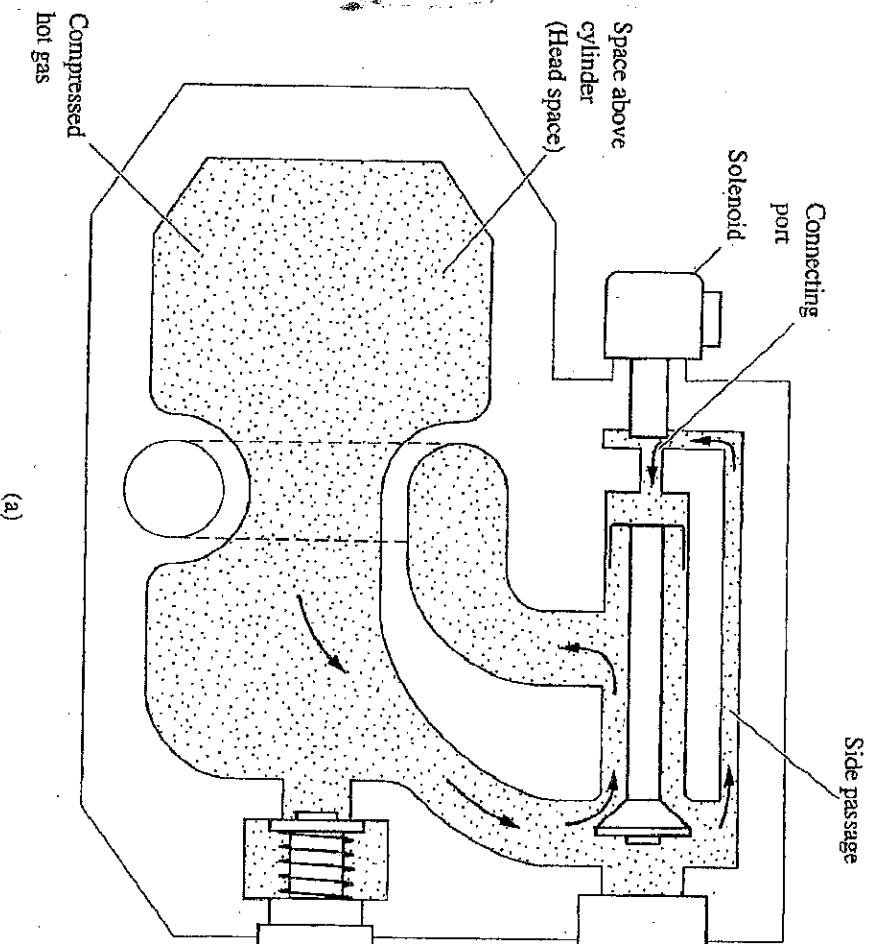
- Variation of discharge air temperature in a packaged unit
- Change in return chilled water temperature in a water chiller
- Changes in suction pressure in the evaporator

Figure 11.13 shows a typical cylinder unloader. When a temperature sensor in a packaged unit detects a drop in the discharge air temperature, the controller energizes the solenoid, draws the plunger away from the connecting port, allows the compressed gas to flow through a side passage, pushes the piston and valve stem of the unloader away from the solenoid, and opens the unloader valve. The compressed gas from the cylinder then flows to the suction side of the compressor. The two cylinders under the same head space are unloaded.

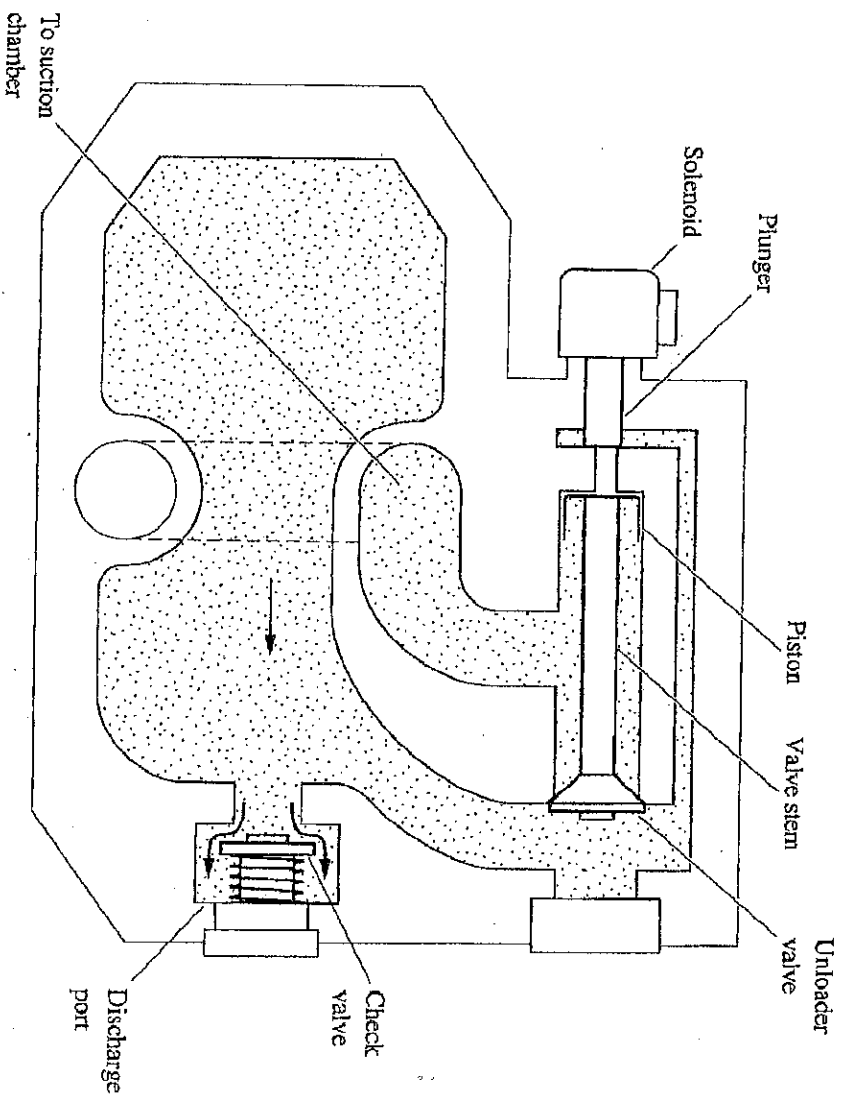
When the discharge air temperature rises, the plunger of the solenoid covers the connecting port. The pressure of the compressed gas exerted on the disk closes the unloader valve. Consequently, the increased pressure in the space above the cylinder opens the check valve and permits the discharge gas to enter the discharge line. The two cylinders are now loaded.

A cylinder unloader is a form of step control. Typically, the capacity of an eight-cylinder reciprocating compressor can be controlled in four steps: 100 percent, 75 percent, 50 percent, and 25 percent.

Speed Modulation. Recent developments in solid-state electronics may allow adjustable-frequency ac inverter-driven variable-speed motors to be used for capacity control in reciprocating compressors. Commercial variable-speed air conditioners with a capacity of few tons of refrigeration were available in Japan in the 1980s.



(a)



(b)

FIGURE 11.13 Cylinder unloader: (a) unloaded cylinders; (b) loaded cylinders.

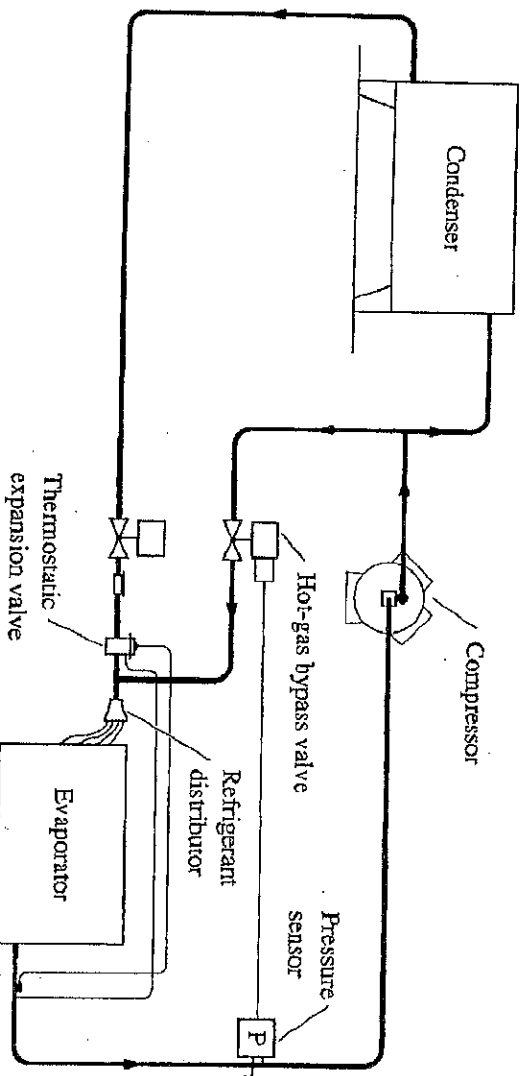


FIGURE 11.14 Hot-gas bypass control for a DX reciprocating system.

Inverter-driven variable-speed reciprocating compressors have the advantage of: lower start/stop or cycling losses, stable temperature control, and better system performance. They also have several drawbacks, such as cooling of the hermetic motor at low refrigerant flow rates, noise at high rotation speeds, and vibration at low speeds. In addition, they require a specially designed expansion valve for low-speed operation because a condensing pressure below a certain limit may affect the normal operation of the expansion valve.

Hot-Gas Bypass Control. In hot-gas bypass control, according to the sensed suction pressure, a certain amount of hot gas that balances the reduction in refrigeration load bypasses the condenser and the expansion valve, and enters the low-pressure side directly. In a DX reciprocating refrigeration system, bypassed hot gas is often introduced between the expansion valve and the liquid refrigerant distributor, as shown in Fig. 11.14.

Although hot-gas bypass control permits modulation of system capacity to a very low percentage, this method does not save energy at reduced refrigeration load. It is often used as a safety device to protect the evaporator or DX coil from frosting or excessive cycling of the compressor during on/off control when the refrigeration load is less than 20 to 25 percent of design capacity, or for industrial applications where strict temperature or humidity control is essential.

ASHRAE/IESNA Standard 90.1-1999 specifies that cooling systems shall not use hot-gas bypass or other evaporator pressure control systems except for (1) systems designed and equipped with multiple steps of unloading or continuous capacity modulation or (2) packaged systems with cooling capacities not greater than 90,000 Btu/h (26 kW). When the cooling system rated capacity is less than or equal to 240,000 Btu/h (70 kW), the maximum hot-gas capacity is 50 percent of the total capacity. When the cooling system rated capacity is greater than 240,000 Btu/h (70 kW), the maximum hot-gas capacity is 25 percent of total capacity.

Safety Controls

Safety controls are used to shut down the refrigeration system during a malfunction and to prevent damage of system components. In reciprocating refrigeration systems, safety controls include low-pressure, high-pressure, low-temperature, frost, oil pressure failure, and motor overload controls.

Low-Pressure and High-Pressure Controls. The purpose of low-pressure control is to stop the compressor when the suction pressure drops below a preset value or when the refrigerant flow rate is too low to cool the compressor motor. Figure 11.15a shows a typical low-pressure control

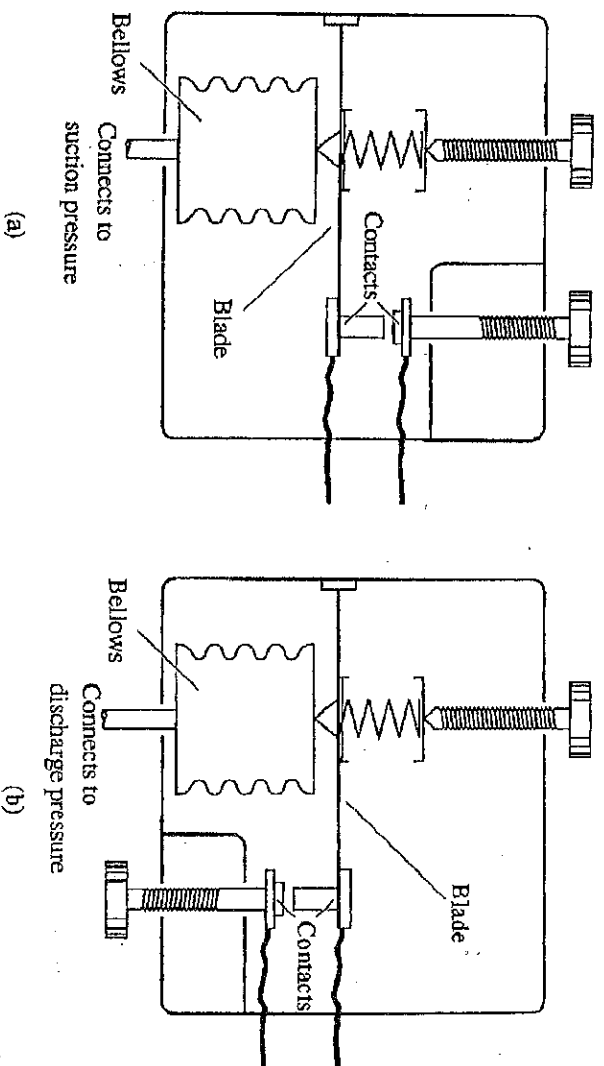


FIGURE 11.15 (a) Low pressure control and (b) high pressure control.

mechanism. When the suction pressure falls below a certain limit, the spring pushes the blade downward, opens the motor circuit, and stops the compressor. When the suction pressure increases, the bellows expand, thus closing the contact of the motor circuit and restarting the compressor. The two adjusting screws are used to set the cut-out and cut-in pressures. Cut-out pressure is the pressure at which the compressor stops, and cut-in pressure is the pressure at which the compressor starts again.

The purpose of high-pressure control is to stop the compressor when the discharge pressure of the hot gas approaches a dangerous level. Figure 11.15b shows a typical high-pressure control mechanism. If the discharge pressure reaches a certain limit, the bellows expand so that the blade opens the motor circuit contact and the compressor stops. When the discharge pressure drops to a safe level, the bellows contract and close the contact, and the compressor starts again. As in a low-pressure control, two adjusting screws are used to set the cut-out and cut-in pressures.

In small refrigeration systems, low-pressure and high-pressure controls are often combined to form a dual-pressure control.

Low-Temperature Control. The purpose of a low-temperature control is to prevent the temperature of chilled water in the liquid cooler from falling below a certain limit, to protect the water from freezing. Freezing water damages the liquid cooler. Figure 11.16 shows a typical low-temperature control system. The sensing bulb senses the chilled water leaving temperature. When the temperature falls below a limit, contraction of the bellows opens the motor circuit and stops the compressor. When the chilled water temperature rises above the limit, the expansion of the bellows closes the circuit and restarts the compressor.

Frost Control. The purpose of frost control is to prevent the formation of frost on the outer surface of the DX coil. Usually, a temperature sensor is used to sense the outer surface temperature of the DX coil. When the temperature drops to 32°F (0°C), the controller actuates a relay, which opens the circuit and stops the compressor.

Oil Pressure Failure Control. The purpose of the oil pressure failure control is to stop the compressor when the oil pressure drops below a certain limit and fails to lubricate the main bearings and other components. Figure 11.17 shows a typical oil pressure failure control mechanism. When

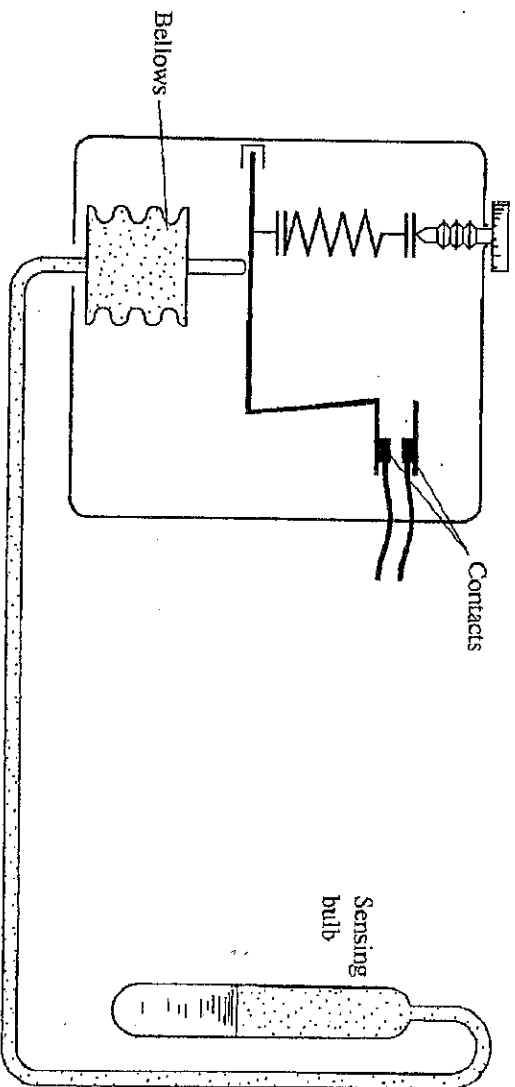


FIGURE 11.16 Low-temperature control.

the compressor starts, points *a* and *b* of the differential pressure switch and points *c* and *d* of the timer switch are in contact with each other. The differential pressure switch is affected by the differential pressure between oil pump discharge pressure and crankcase suction pressure. If the pressure of oil discharged from the oil pump does not reach a predetermined level within a certain time interval, the low oil pressure keeps the differential pressure switch *a* and *b* in a close contact position.

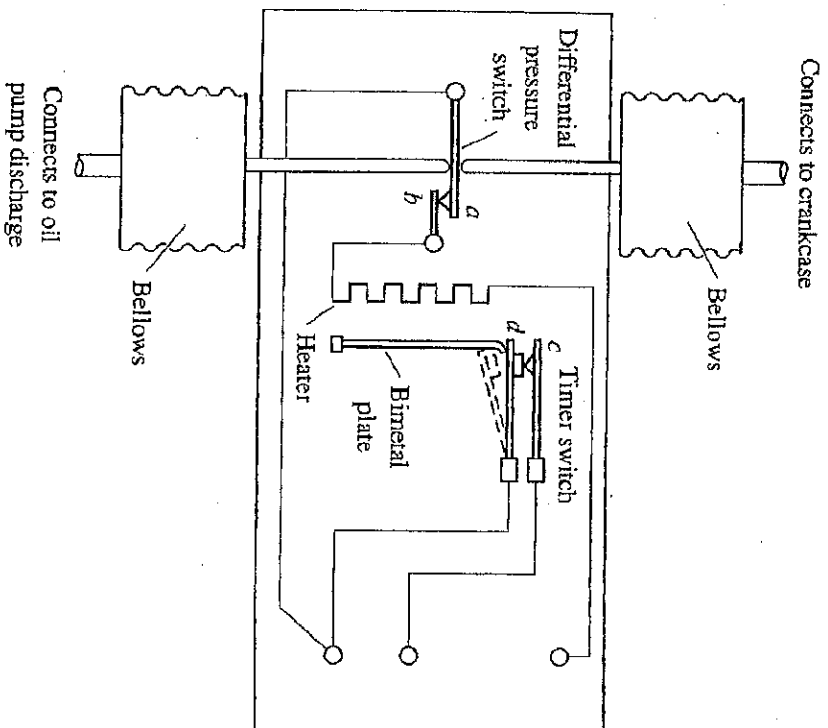


FIGURE 11.17 Oil pressure failure control.

The heater is energized, which causes the bimetal plate to deviate from the vertical position. The bimetal plate no longer keeps the timer switch d in contact with c , and the compressor stops.

If the oil pressure reaches the preset value within the predetermined time interval, a does not make contact with b . The breaking in contact of a and b deenergizes the heater before the bimetal plate starts to bend. The bimetal plate still holds the timer switch d in contact with c , and the compressor continues to operate.

If the oil pressure drops below the preset value and does not rise again within a certain time interval, the energized heater causes the bimetal plate to bend, opens the contact of the timer switch, and stops the compressor.

Motor Overload Control. The purpose of motor overload control is to protect the motor from dangerous overheating caused by a long period of overloading or failure of the compressor to start. Thermal protectors with bimetal or thermistor sensors are installed inside the motor or mounted on the outer surface of the motor shell. These sensors sense the temperature or both the temperature and electric current supplied to the motor windings. If dangerous overheating and overloading occur, a controller breaks the circuit.

Recently developed microprocessor-operated DDC system controllers combine all safety control mechanisms with other control systems of a reciprocating refrigeration system in one package. The DDC system controller senses, analyzes, and operates the safety control systems in coordination with other systems with far more functions and more sophisticated comprehensive controls.

11.6 PERFORMANCE OF RECIPROCATING COMPRESSOR, EVAPORATOR, AND CONDENSER

Performance of Reciprocating Compressor

There are two important parameters in the performance analysis of a reciprocating compressor: refrigeration capacity Q_{ev} , Btu/h or tons (W or kW) of refrigeration (as described in Sec. 9.10); and power input P_{in} , hp (kW); or energy efficiency ratio EER, Btu/h (W) (as defined in Sec. 9.20). Because Q_r and P_{in} are always affected by suction pressure P_{suc} and discharge pressure P_{con} (both in psig), it is convenient to relate the performance of the reciprocating compressor to suction temperature T_{suc} (in °F) and to the condensing temperature T_{con} . The saturated pressure at T_{ev} is equal to P_{ev} . Suction pressure is approximately equal to the saturated pressure of $T_{ev} - 2$, and discharge pressure is approximately equal to the saturated pressure of $T_{con} + 1$, both in °F (°C).

Consider an accessible hermetic compressor with the following operating and construction characteristics:

Refrigerant	HFC-22
Number of cylinders	8
Bore of cylinder	2.65 in. (67.3 mm)
Stroke of piston	2.25 in. (57.2 mm)
Revolutions per minute	1750
Suction temperature	40 °F (4.4 °C)
Condensing temperature	115 °F (46.1 °C)
Subcooling	10 °F (5.6 °C)
Superheating	10 °F (5.6 °C)

aldehydes and organic acids. Potassium permanganate—impregnated alumina (chemisorbers are discussed in Sec. 24.5) is effective against sulfur and nitric oxides, hydrogen sulfide, and low-molecular-weight aldehydes.

15.18 HUMIDIFICATION AND HUMIDIFIERS

Space Relative Humidity

As mentioned in Sec. 4.9, for comfort air conditioning systems the space relative humidity is tolerable between 35 and 65 percent in summer, preferably from 40 to 50 percent. During winter, for space served by a comfort air conditioning system that is installed with a humidifier, the space relative humidity should not exceed 30 percent except in hospitals. The actual humidifying capacity of a humidifier should not exceed the humidifying requirement so that wet surfaces do not occur inside the AHU, PU, or supply ducts. Wet surfaces and dirt often cause the growth of microorganisms and thus indoor air quality problems.

For air conditioning systems without humidifiers, space relative humidity is usually not specified.

For processing air conditioning systems, the space relative humidity should be specified as required by the manufacturing process.

If the space temperature is maintained at 72°F (22.2°C) and only free convection is provided in the occupied space, the space relative humidity should not exceed 27 percent when the outdoor temperature is 30°F (−1.1°C), in order to prevent condensation on the inner surface of single-glazed windows during winter. To prevent condensation on the inner surface of double-glazed windows, the space relative humidity should not exceed 33 percent when the outdoor temperature is 0°F (−17.8°C).

Humidifiers

A humidifier adds moisture to the air. Humidifiers may (1) inject steam directly into air or add heat and evaporate steam from water supplied to the conditioned space; (2) atomize or spray liquid water, so that water evaporates and is added to the air; or (3) force air to flow over a wetted element so that as water evaporates, it is added to the air as vapor. All these increase the humidity ratio of the space air and, therefore, its relative humidity. One important index of a humidifier is its humidifying capacity $\dot{m}_{\text{cap}}$ in pounds of water per hour, lb/h (kg/h), or the rate at which water vapor is added to the air.

Humidifying Load

Humidifying load $\dot{m}_{\text{hu}}$, in lb/h or kg/h, is the amount of water vapor required to be added to the air by a humidifier so as to maintain a predetermined space relative humidity. The humidifying load of an air system installed with a humidifier can be calculated as (see Fig. 15.41)

$$\dot{m}_{\text{hu}} = 60\dot{V}_s(\rho_s(w_{\text{in}} - w_{\text{en}})) = 60\dot{V}_s\rho_s(w_s - w_m) \quad (15.52)$$

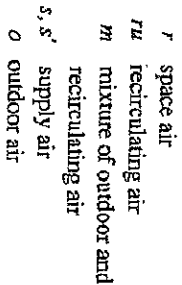
where $\dot{V}_s$ = supply volume flow rate of AHU or PU, cfm (m³/min)

ρ_s = density of supply air, lb/ft³ (kg/m³)

$w_{\text{en}}, w_{\text{in}}$ = humidity ratio of air entering and leaving humidifier, lb/lb (kg/kg)

w_s = humidity ratio of supply air, lb/lb (kg/kg)

w_m = humidity ratio of mixture of outdoor air and recirculating air, lb/lb (kg/kg)



Humidifying load for a typical air system.

In Fig. 15.41, s is the state of supply air when the space has a cooling load, $w_s = w_{s'}$.

The humidifying load can also be calculated from the following relationship:

(15.53)

where $\dot{V}_o$ = supply volume flow rate of outdoor air intake, cfm (m^3/min)

V_{inf} = volume flow rate of infiltrated air, cfm (m^3/min)

w, w_o = humidity ratio of space air and outdoor air, lb/lb (kg/kg)

In Eq. (15.53), m_{mv} represents the space moisture gains, in lb/h (kg/h). Space moisture gains include the latent load from the occupants, appliances, equipment, and products. The moisture gains from the building structures are often ignored.

Types of Humidifier

According to the mechanism used for evaporation of water vapor from water, humidifiers can be classified as steam and heating element humidifiers, atomizing humidifiers, and wetted element humidifiers.

15.19 STEAM AND HEATING ELEMENT HUMIDIFIERS

At a low partial pressure when mixed with dry air, steam is the ready-made water vapor. With proper water treatment at the boiler, steam is free of mineral dust and odor, and it does not support the growth of bacteria that create sanitation problems. Two types of steam humidifiers are currently used in air systems: steam grid humidifiers and steam humidifiers with separators.

Steam Grid Humidifiers

A steam grid humidifier installed inside ductwork is shown in Fig. 15.42a. A steam grid humidifier may have a single distribution manifold or multiple manifolds. Each distribution manifold has an

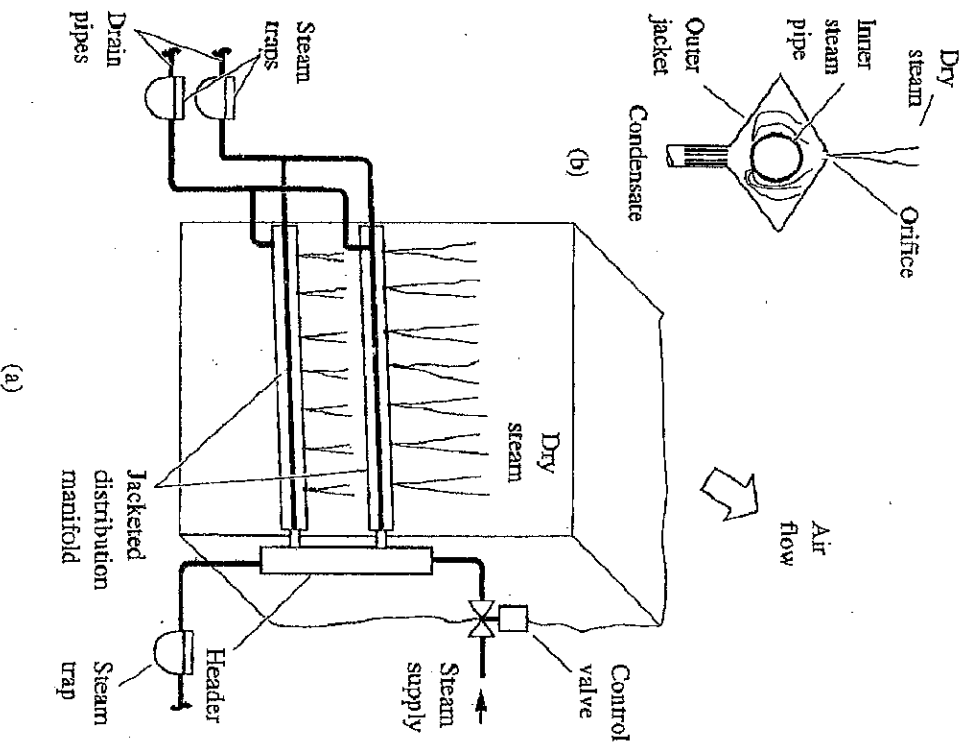


FIGURE 15.42 Steam grid humidifiers: (a) steam grid and piping connections; (b) cross section of jacketed distribution manifold.

inner steam pipe and an outer jacket. The inner steam tube is connected to a header and controlled by a control valve. Steam is supplied at a pressure less than 10 psig (69 kPag) and throttled to a lower pressure after the controlled valve. It then enters the inner tube and discharges through the small holes to the outer jacket. The dry steam is again discharged through the orifices of the jacket to the ambient airstreams to humidify them, as shown in Fig. 15.42b. The condensate inside the jacket is discharged to a drainpipe and steam trap located at the opposite side. The inner tube and the outer jacket are slightly pitched toward the drainpipe and steam trap.

Steam Humidifiers with Separators

A steam humidifier with a separator is illustrated in Fig. 15.43. Steam is supplied to a jacketed distribution manifold and then enters a separating chamber with its condensate. It then flows through a control valve, is throttled to slightly above atmospheric pressure, and enters a drying chamber. Because of the lower pressure and temperature in the drying chamber compared with the higher pressure and temperature in the surrounded separating chamber, the steam is superheated. Dry steam is then discharged into the ambient airstream through the orifices of the inner steam discharging tubes. Noise is produced mainly during the throttling of the steam at the control valve. It is attenuated as steam flows through the silencing materials.

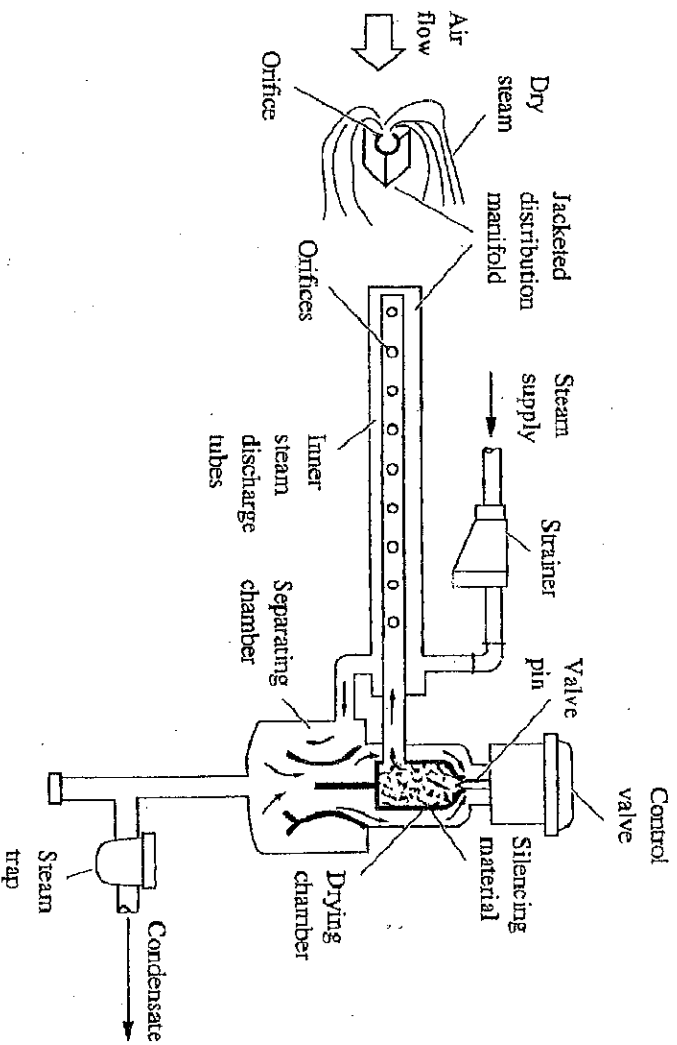


FIGURE 15.43 Steam humidifier with a separator. (Adapted with permission from *The Armstrong Humidification Handbook*, 1991.)

Heating Element Humidifiers

When a steam boiler is not available or is too expensive to install, steam humidification is still possible by using heating element humidifiers. A heating element humidifier has a water pan with electric or gas-fired heated elements installed at the bottom of the pan to evaporate the liquid water to water vapor and then add the vapor to the airstream flowing over the water pan, as shown in Fig. 15.44. The humidifying capacity of a heated element humidifier is limited by its heat-transfer surface and the characteristics of the heating element.

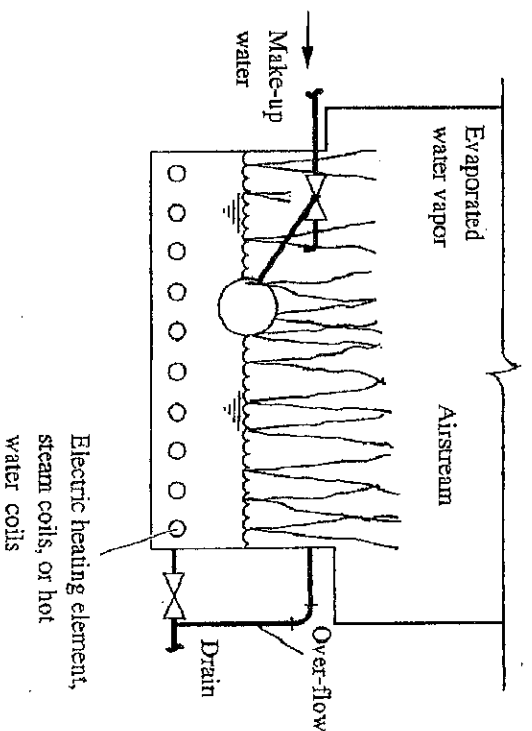


FIGURE 15.44 Heating element humidifier.

Characteristics and Requirements of Steam and Heating Element Humidifiers

Steam humidifiers are widely used in commercial and industrial applications. Both steam and heating element humidifiers need heat energy from gas, fossil fuel, or electricity to evaporate the liquid water in the steam generator. The temperature of the ambient air or airstream in which moisture has been added is approximately unchanged. Steam and heating element humidifiers are therefore often called isothermal humidifiers. For a conditioned space that has a cooling load year round and needs a cold air supply in winter, it is a waste of energy to evaporate the liquid water by using fossil fuel instead of the excess heat gains at the conditioned space.

Control is essential in steam and heating element humidifiers. The capacity of a steam humidifier is often controlled by a microprocessor-based controller by modulating the valve pin of the control valve according to the signal of a humidity sensor and, therefore, the steam flow rate. Morton (1996) reported an accuracy of humidity control of ± 5 to 7 percent for on/off control and ± 3 to 5 percent for modulation control. The size of the control valve should be carefully matched with the humidifying load with an adequate turndown ratio. The control valve should be integrated with the humidifier and steam-jacketed at supply pressure to prevent condensation. An interlocking control should be installed to drain all condensate before the steam humidifier is started.

For an electric heating element humidifier, or simply an electric humidifier, the input wattage can be adjusted when the space relative humidity is too high or too low. A gas-fired heating element humidifier, or simply a gas-fired humidifier, uses infrared burners located within a heat exchanger. A microprocessor-based controller stages and cycles the burner to vary the heat output and therefore the humidifying capacity during part-load operation.

The makeup water supply to the steam boiler and the water pan must be well treated so that the steam is free of mineral deposits and odor. For electric and gas-fired humidifiers, according to DeBat (1996), recent developments, such as ionic bed technology, overcome the problem of mineral deposits. Dissolved mineral salts begin to precipitate out of water when their concentration becomes too high. Ionic beds consist of inert fibers that attract these precipitated minerals. The mineral salts deposit on the ionic bed instead of on the walls of the heating chamber. When thoroughly encrusted with crystalline solids, the ionic bed cartridge is easily removed and discarded.

Steam humidifiers should be located where noise is not objectionable. Access to equipment for inspection and maintenance should be provided. Condensate drainage must be properly designed. If the steam humidifier is located in ductwork, install a duct high-limit humidistat and an interlocked airflow proving switch. The humidity sensor is located preferably in the return air duct or a representative location.

15.20 ATOMIZING AND WETTED ELEMENT HUMIDIFIERS

Humidification Process

Atomizing means producing a fine spray. When liquid water is atomized, the smaller the diameter of the water droplets, the greater the interfacial area between water and air and thus the higher the rate of evaporation and humidification.

When air flows through an atomized water spray from an atomizing humidifier, if the difference between the temperature of the water spray and the wet-bulb temperature of the ambient air is small, the result is an increase in the humidity ratio of air from the addition of evaporated water vapor and a corresponding drop in air temperature because of the absorption of required latent heat of vaporization from the ambient air. Such a humidification process is an adiabatic saturation or evaporative cooling process and follows the thermodynamic wet-bulb temperature line on the psychrometric chart.

Atomizing humidifiers can be classified, according to their configuration and mechanism of atomizing, as ultrasonic humidifiers, centrifugal atomizing humidifiers, pneumatic atomizing humidifiers, and air washers.

Ultrasonic Humidifiers

An ultrasonic humidifier uses a piezoelectric transducer to convert high-frequency mechanical electric signals to a high-frequency oscillation. The transducer is submerged into a water pan in which water is well treated and demineralized. The negative oscillation produces a momentary vacuum that causes the water to cavitate into vapor at low pressure. The positive oscillation creates a high-compression wave which forces the water particles to leave the water surface and inject into the airstream. Water films are then broken into a fine mist. Air containing this mist can be blown into the space directly or through an air system by a fan.

If the water is well treated, it will be free of mineral deposits, and no filter is required downstream from the humidifier. Ultrasonic humidifiers create less equipment noise than other atomizing humidifiers. In the mid-1990s, ultrasonic humidification had been installed in telecommunication projects. Ultrasonic atomizing humidifiers with small humidifying capacities have been used directly in residences.

Case Study: White Plains Ultrasonic Humidification Project

Longo (1994) reported on a retrofit ultrasonic humidification project which is located in a 1950s building in White Plains, New York. The humidification system in a totally new air system serves space that houses solid-state complex multistwitching processors. Static charges created in a low-humidity environment can be detrimental to system components and cause failure.

The 100,000-cfm (47,200-L/s) air system consists of four supply fans (two standby), high-efficiency filters, and cooling coils with outside air economizer for free cooling. The ultrasonic humidifiers were placed after the filters and upstream from the supply fan. There are altogether 43 ultrasonic units mounted on stainless-steel racks, each rated at 39.6 lb/h (18 kg/h) and 960 W. A humidity sensor is installed in the return air duct. The controller turns the humidity system on one stage at a time up to a total of six stages. A high-limit sensor set at 80 percent relative humidity shuts down the system in case of overhumidification.

Water is treated by a series of prefilters, a reverse osmosis system, and deionization canisters. Water quality is controlled by a resistivity probe. When water resistance reaches 0.02 M Ω , the deionization canisters will be changed.

After completion, the actual running results in winter are excellent. As the equipment was installed gradually, the supply air temperature was rescheduled, and the space relative humidity was maintained precisely at 50 percent. The initial cost of the White Plains humidity system was \$350,000. Annual operating cost was \$23,000, which was less than one-tenth of the annual operating cost of an electric heating element humidity system.

Centrifugal Atomizing Humidifiers

These humidifiers use the centrifugal force produced by a rotating device such as a rotating cone, blades of an axial fan, rotating disk, or rotating drum to break the liquid water film into fine mist or to fling it into fine water droplets. When air is forced through an atomizing water spray, produced by a pulverizing fan, water vapor is evaporated and added to air as a result of the mass concentration difference between the saturated air film at the surface of the droplets and the ambient air. The rotating device receives liquid water either from a pressurized water supply or by dipping into the surface of a nonpressurized supply. The humidifying capacity of a pulverizing fan depends mainly on the volume flow rate of the fan. Because of the oversaturation characteristics of the pulverizing fan, $\dot{m}_{\text{cap}}$ may vary from 50 to 150 lb/h (23 to 68 kg/h). Oversaturation is the excess amount of water particles present in the moist air, and it is discussed in Chap. 20. Many different types of rotating humidifiers with limiting humidifying capacities are used directly in the conditioning space in residential and industrial applications. Most are portable.

Maintaining a predetermined space relative humidity with a centrifugal atomizing humidifier can be performed by modulating the flow rate of water supplied to the rotating devices.

Pneumatic Atomizing Humidifiers

A typical pneumatic atomizing nozzle consists of two concentric brass tubes, as shown in Fig. 15.45*a*. The inner tube has an outer diameter of $\frac{3}{32}$ in. (2.4 mm) and an inner diameter of $\frac{1}{32}$ in. (0.8 mm). It is connected to a water tank that can be moved to adjust the difference in water levels between the water tank and the centerline of the nozzle. The conical outer tube has a minimum inner diameter of $\frac{1}{8}$ in. (3.2 mm) and is connected to a compressed air line with a pressure at 15 psig (103 kPag).

When compressed air is discharged from the annular slot at a very high velocity, it extracts water from the inner tube and breaks the water into a very fine mist. A typical pneumatic atomizing nozzle may have a humidifying capacity $\dot{m}_{\text{cap}}$ from 6 to 10 lb/h (2.7 to 7.5 kg/h). The magnitude of $\dot{m}_{\text{cap}}$ is affected by the configuration of the nozzle, the pressure of the compressed air, and the difference in water levels. In a pneumatic humidifier, $\dot{m}_{\text{cap}}$ is usually controlled by adjusting the difference in water levels between the water tank and the centerline of the nozzle.

A pneumatic humidifier produces high-frequency noise. This type of humidifier is usually applied in industries with a certain sound level of machine noise and can be used for direct in-space humidification.

Wetted Element Humidifiers

Wetted element humidifiers include a wetted element, such as an evaporative pad, plastic, or impregnated cellulose, that is dipped with water from the top. Such humidifiers have been installed in air-handling units and packaged units to humidify the air. Characteristics of wetted element humidifiers are similar to those of wetted element evaporative coolers (see Chap. 27). Modulation of the water supply to the water dipping device varies the humidifying capacity and maintains the desirable space relative humidity.

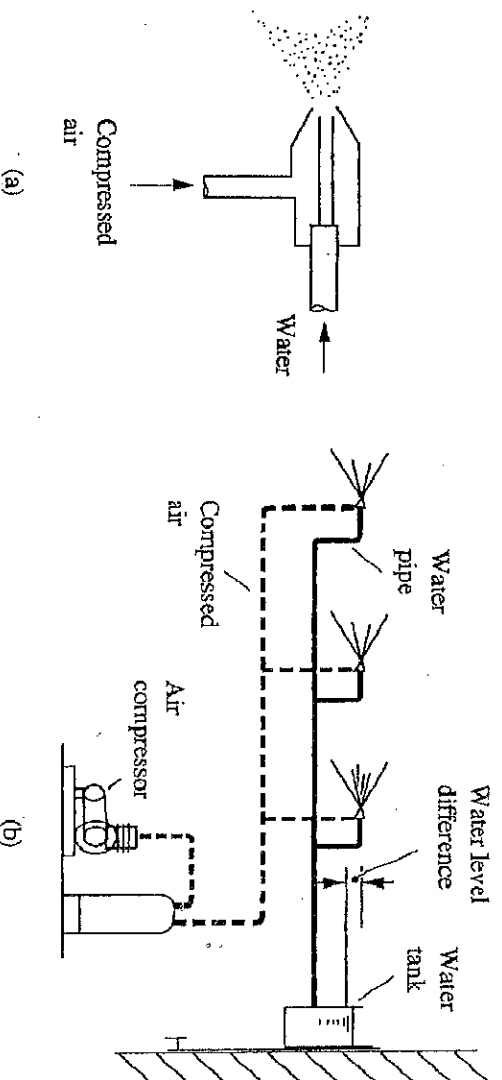


FIGURE 15.45 Pneumatic atomizing humidifier (a) nozzle; (b) pneumatic atomizing humidifying system.

For a cooling and dehumidification process, the performance of an air washer is better described by a factor called the *performance factor* F_p as follows:

$$F_p = \frac{h_{en} - h_N}{h_{en} - h_s^*} \quad (15.55)$$

where h_{en}, h_N = enthalpy of air entering and leaving air washer, Btu/lb (kJ/kg)

h_s^* = enthalpy of saturated air at chilled water spraying temperature, Btu/lb (kJ/kg)

The performance of an air washer for a humidification process, indicated by the saturation efficiency ϵ_{sat} and for a cooling and dehumidification process, expressed by F_p , depends mainly on the contact time between air and water and the contact surface area, i.e., the water-air ratio $\dot{m}_w/\dot{m}_a$ (which indicates the mass flow rate of the spraying water to the mass flow rate of the air flowing through the air washer, in same units), the length of the air washer, and the direction of spray with respect to airflow, (whether it is opposing or following the airflow).

An air washer is usually designed at an air velocity between 500 and 800 fpm (2.5 and 4 m/s) with respect to its cross-sectional area at the water sprayers. Its greater washer length and the attached water tank result in a bulkier volume than that of DX coils and water cooling coils. Total pressure loss of the airstream flowing through an air washer $\Delta p_{t,w}$, in in. WC (Pa), depends mainly on the configuration of the eliminators and the air velocity flowing through them. Usually $\Delta p_{t,w}$ varies from 0.25 to 1 in. WC (62.5 to 250 Pa), and typically it is 0.5 in. WC (125 Pa). For humidification and evaporative cooling, the water-air ratio $\dot{m}_w/\dot{m}_a$ is usually 0.3 to 0.6. For a 5-ft (1.5-m) air washer with $\dot{m}_w/\dot{m}_a = 0.45$, that is, 4 gpm of water per 1000 cfm of air (0.534 L/s of water per 1000 L/s of air), $\epsilon_{sat} = 0.85$ to 0.9. For cooling and dehumidification, the water-air ratio is usually 0.5 to 1.2. For a 5-ft (1.5-m)-long air washer with $\dot{m}_w/\dot{m}_a = 0.9$, that is, 8 gpm/1000 cfm (1 L/s of water per 1000 L/s of air), $F_p = 0.5$ to 0.75.

For either the humidification process or the cooling and dehumidification process in an air washer, oversaturation always exists at the exit. Oversaturation ($w_o - w_s^*$) usually varies between 0.0002 and 0.001 lb/lb (0.0002 and 0.001 kg/kg). It depends mainly on the construction of the eliminator and the flow-through air velocity. Oversaturation is beneficial to a humidification process with space heat gains and interferes with a cooling and dehumidification process. Because of the existence of oversaturation, the saturation effectiveness is no longer the dominant index that affects the performance of the air washer during humidification. Oversaturation is discussed in Sec. 20.13.

Bypass Control

Bypass control is defined as modulating the face damper and the bypass damper of the air washer to vary the proportion of humidifying air and bypass air in order to maintain the space relative humidity within predetermined limits. Such a control is often used in an air washer to provide better humidity control than modulating the water pressure or the water flow rate to the spraying nozzles.

Single-Stage or Multistage

A group of nozzles that are supplied with water at the same pressure and produce spray at the same temperature are said to be in the same stage. A stage may have one or two spraying banks. A two-stage air washer is illustrated in Fig. 15.47.

For humidification processes or for cooling and dehumidification processes using chilled water from the refrigeration plant, it is more economical to operate a single-stage air washer, usually with two water-spraying banks. When low-temperature deep well or groundwater is used for cooling and dehumidification, it is often advantageous to use two-stage, or even three-stage, water spraying to make full use of the cooling capacity of the groundwater.

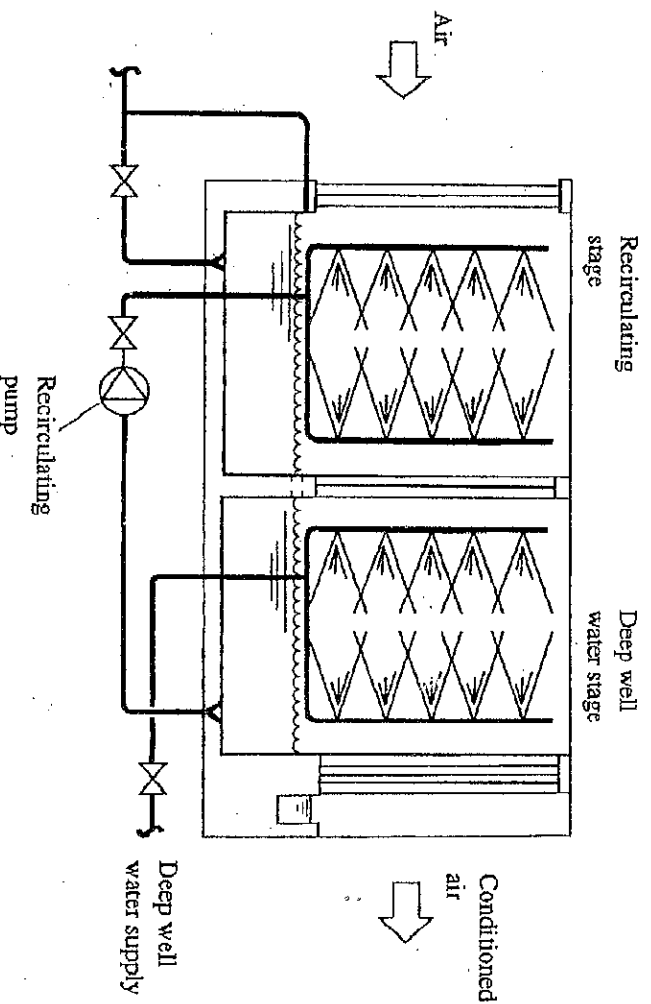


FIGURE 15.47 Two-stage air washer using deep well water.

Groundwater can be used again as condenser water, or as flushing water in plumbing systems after its discharge from the air washer following necessary water treatment. Used groundwater should be drained to another well (recharge well) in the vicinity to prevent subsidence and damage to the soil, which may affect the foundation of the building in the vicinity of the deep well.

The use of groundwater is regulated by local and federal environmental protection agencies.

15.22 CHARACTERISTICS OF ATOMIZING AND WETTED ELEMENT HUMIDIFIERS

Atomizing and wetted element humidifiers, including air washers for humidification purposes have the following operating characteristics:

- Conditioned air follows the thermodynamic wet-bulb temperature line on a psychrometric chart. The humidity ratio of the air increases while it is evaporatively cooled. Atomizing humidifiers are often called nonisothermal humidifiers. This characteristic indicates that atomizing and wetted element humidifiers are especially suitable for conditioned spaces that need a cold air supply to offset a space cooling load during or after the humidification process. The heat energy required to evaporate the liquid water, as in steam or heating element humidifiers, has been saved. For a conditioned space which needs a warm air supply during winter operation, the temperature of evaporatively cooled air after an atomizing or wetted element humidification process should be raised before supplying it to the conditioned space, so as to offset the transmission loss through the building envelope. The heat energy required is approximately equal to that required to evaporate water in a steam boiler or in a water pan when a steam humidifier or a heating element humidifier is used.
- For atomizing humidifiers that need a cold air supply during winter, when the humidifying capacity is modulated, both the space relative humidity and temperature vary accordingly. Many

- industrial manufacturing processes need to maintain a stable space relative humidity (RH). RH is often the controlled variable, and temperature is usually maintained within predetermined limits.
- Oversaturation exists at the exit of the atomizing humidifier even if an eliminator is used. Oversaturation is often advantageous for humidifiers installed in conditioned spaces having excess heat gains because it increases the humidifying capacity.

- The size of the water droplets, the local humidifying capacity, and the excessive heat gains determine whether liquid water droplets may fall on the floor, products, or equipment. According to field measurements, the diameter of water droplets suspended in the airstream discharged from atomizing humidifiers is as follows:

Pneumatic atomizing humidifiers	30 to 80 μm
Centrifugal atomizing humidifiers	50 to 300 μm

When the diameter of the water droplets exceeds 80 μm and the local humidifying capacity is excessive, water may drip. Water dripping must be avoided, even if reducing the humidifying capacity is necessary. It causes product damage and corrosion of appliances, instruments, and equipment.

- In the water tank and water pan of atomizing and wetted element humidifiers, algae, bacteria, and other microorganisms may grow. Water spray and the airstream discharged from the atomizing and wetted element humidifiers may distribute the bacteria and sometimes odors. Water treatments, regular cleaning and flushing, and periodic inspection and blowdown help to minimize these problems. The designer must include convenient access for such maintenance.

15.23 SELECTION OF HUMIDIFIERS AND DESIGN

The following factors should be considered in selecting a humidifier:

- *Energy consumption and operating cost.* For a space that needs a cold air supply even in winter, atomizing and wetted element humidifiers save energy and, therefore, have lower operating costs. Among the atomizing humidifiers, pneumatic atomizing humidifiers require compressed air to atomize the liquid water and, therefore, have a higher operating cost than other atomizing humidifiers.
- *Quality of humidification.* Humidified air should be clean and free of odor, bacteria, particulate matter, and water droplets to prevent microbial growth due to wetted surfaces or standing water and improving IAQ.
- *Humidifying capacity.* Is a large, medium, or small humidifying capacity required? Is oversaturation beneficial in the conditioned space?
- *Capacity control.* Is this an on/off control or a modulation control, i.e., a stepless control? Does the control have a fast or slow response? What is the turndown ratio of the control valve?
- Morton (1996) recommended that the best location of a humidifier in an AHU or PU is in the first section of supply main duct downstream from the supply fan. The next choice is between the coil section and the supply fan if there is no oversaturation in the airstream.
- *Equipment noise.* For in-space humidifiers such as pneumatic humidifiers, equipment noise is a primary concern.
- *Initial cost.* For a steam humidifier, the initial cost should also include the installation costs of the steam boiler and the corresponding steam piping system.
- *Maintenance.* This includes the amount and cost of required maintenance work for the humidifiers.
- *Space occupied.* This is the volume occupied by the humidifier per unit humidifying capacity.

TABLE 15.4 Comparison of Various Types of Humidifiers

Description	Steam	Heating element	Air washer	Ultrasonic	Pneumatic atomizing
Humidifying capacity	Small to large	Small	Large	Small to large	Medium and large
Oversaturation	No	No	Yes	Possible	Possible
Quality of humidification	Good	Good	Good	Good	Good
Dripping of water in conditioned space	No	No	No	No	No
Energy consumption and operation cost	Need more energy if cold air supply is needed, so operating cost is higher	Same as steam	Saving energy if cold air supply is needed in winter	Saving energy if cold air supply is needed in winter	Saving energy if cold air supply is needed in winter
Initial cost	High	High	Medium	Medium	High
Maintenance work and maintenance cost	Low	Medium	High	Medium	Medium
Capacity control	Modulation	On/off or modulation	Modulation	Multiple stages	Modulation
Response to control	Fast	Slow	Medium	Fast	Fast
Space occupied	Small	Medium	Large	Small	Medium
Equipment noise	Medium	Low	Medium	Low	High

Various types of humidifiers are compared in Table 15.4. For commercial buildings that need a warm air supply during winter and for which a steam supply is available, a steam humidifier is often the most suitable choice. For a small packaged unit serving a health care facility, a heating element humidifier may be selected. For a telecommunications center with an appreciable amount of space heat gain, an ultrasonic humidifier is often the right choice. Ultrasonic humidifiers are also often used in residences for their comparatively low equipment noise. For a factory in which space air contains lint and other dirt that should be cleaned, or for which a cold air supply is needed to offset the cooling load in winter, an air washer may be the suitable choice.

When air washers or atomizing humidifiers are selected as the humidifiers, oversaturation may result in the supply air discharged from the air washer or atomizing humidifier. As oversaturation causes wetted surfaces and dampness along the air distribution passage, the following remedies are recommended to prevent IAQ problems:

- All-metal (fiber-free) silencer or active silencer should be used, if required.
- Interior surfaces of ducts that contact supply air must be galvanized metal sheets.
- Duct heat gain may be beneficial to eliminate oversaturation.
- Periodic monitoring of the moisture and dirt conditions of the ducts and silencers should be provided.

ASHRAE/IESNA Standard 90.1-1999 mandates that when a zone is served by a system or systems with both humidification and dehumidification, means (such as limit switches, mechanical stops, software program for DDC control) shall be provided to prevent simultaneous operation of humidification and dehumidification equipment. Exceptions are zones served by desiccant dehumidification systems, which are used with direct evaporative cooling in series, or zones where specific humidity levels are required such as computer rooms, museums, and hospitals. Standard 90.1-1999 also mandates that humidifiers with preheating jackets mounted in the airstream shall have an automatic valve to shut off preheat operation when humidification is not in use.

